



EMAG Maschinenfabrik GmbH

Austraße 24
73084 Salach - Germany
Phone: +49 7162 17-0
Fax: +49 7162 17-4270
E-Mail: info@emag.com
Internet: www.emag.com

EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

HYUNDAI WIA Corporation
Mr. Kyung Yang Lee
153, Jeongdong-ro,
Seongsan-gu,
Changwon-si,
Gyeongsangnam-do 51533
SOUTH KOREA

Offer

Offers- No: 196563 / 5
date: 19/08/2022

Your technology contact

Andreas Liebrich

Phone +49 7162 17-4693
E-mail aliebrich@emag.com

Your contact

Do-Youn Koo

Phone +82 31 766-4415
E-mail dykoo@emag.com

Your Reference

Inquiry dated 29/11/2021
Contact Kyung Yang Lee
Phone +82 55 2809501
E-mail ardanji@hyundai-wia.com

EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 DUO KBU for straight ball grooves (AC- and CUJ-joints) for the simultaneous soft turning and milling and two Outer Races of the same type

Outer Race BJ#22
Outer Race BJ#23
Outer Race CUJ#27



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Price does not include options (*)

Offer : 196563 / 5

Managing Directors:
Dr. Guido Hegener, Dr. Mathias Klein
Commercial Register Ulm: HRB 533796
Tax No.: 63002/12866 FA Göppingen
VAT No.: DE 813 898 840

HYUNDAI WIA Corporation

Bank accounts:
UniCredit Bank AG / Hypovereinsbank Stuttgart 'IBAN DE02 6002 0290 0322 6388 17' BIC HYVEDEMM473
Commerzbank AG Tübingen 'IBAN DE09 6414 0036 0890 9103 00' BIC COBADEFF641
Kreissparkasse Göppingen 'IBAN DE96 6105 0000 0015 5885 93' BIC GOPSDE6GXXX
Bayerische Landesbank München 'IBAN DE59 7005 0000 0004 2569 67' BIC BYLADEMMXXX

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Price Summary

1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 DUO KBU

1.1 BASIC MACHINE

1.1.14	1	PRICE OF MACHINE VSC 315 DUO KBU (SIEMENS) - SOFT MACHINING	734.700,00 €
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1.2 ADDITIONAL EQUIPMENT

1.2.2	4	TOOL BLOCK FOR 1 TURNING TOOL CAPTO C6 manually clamped	61.600,00 €
1.2.4	1	RAW-PART STORAGE UNIT (SUPPLIED BY CUSTOMER)	0,00 €
1.2.5	1	LINEAR GANTRY LP3, WITH ROTATION MODULE	23.300,00 €
1.2.6	1	GRIPPER JAWS OUTER RACE CLAMPING DIA. 20-36	0,00 €
1.2.7	1	POSITION DETECTION	2.550,00 €
1.2.8	1	NIO - CHUTE, LENGTH 1550mm, R-L / L-R	3.300,00 €
1.2.9	1	SHUTTLE ZR120DUO L-R/ L-L FOR WORKPIECE LOADING	40.700,00 €
1.2.10	2	WORKPIECE CARRIER EXPANDABLE OUTER RACE BJ#22	7.900,00 €
1.2.11	2	WORKPIECE CARRIER EXPANDABLE OUTER RACE BJ#23	7.900,00 €
1.2.12	2	WORKPIECE CARRIER EXPANDABLE OUTER RACE CUJ#27	7.900,00 €
1.2.13	2	AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK	2.400,00 €
1.2.14	1	WORKPIECE UNLOADING by motorised, chain-type conveyor	28.500,00 €
1.2.15	1	LIGHT BEAM FOR UNLOADING CONVEYOR	900,00 €
1.2.16	1	SAFETY ENCLOSURE	12.100,00 €
1.2.18	1	2 SWARF CONVEYORS, CHAIN TYPE	27.400,00 €
1.2.19	2	CENTRAL EXTRACTOR UNIT (PREPARATION)	1.650,00 €
1.2.20	1	CHIP BOX (LARGE VERSION)	2.750,00 €
1.2.21	1	AIR GUN	1.050,00 €
1.2.23	1	NC-GUIDED EMERGENCY RETRACTION AT POWER FAILURE (ESR)	10.500,00 €
1.2.24	1	NETWORK DRIVE CONNECTION	2.000,00 €
1.2.25	2	INTERFACE TO EXTERNAL HANDLING SYSTEM - HARDWARE	7.400,00 €
1.2.26	1	MINI MANUAL CONTROL UNIT (SIEMENS)	4.500,00 €
1.2.27	1	CONNECTING KIT FOR MINI HAND-HELD CONTROL UNIT – SIEMENS	825,00 €

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1.2.28	1	WORKSHOP-ORIENTATED PROGRAMMING AC-JOINTS - SOFT MACHINING	6.000,00 €
1.2.29	1	WORKSHOP-ORIENTATED PROGRAMMING CUJ-JOINTS - SOFT MACHINING	6.000,00 €
1.2.31	1	SERIES TRANSFORMER - MEXICO	3.750,00 €
1.2.32	1	AUXILIARY COOLING SYSTEM	3.500,00 €
1.2.34	1	4-COLOUR STATUS LIGHT (Q-LIGHT)	1.200,00 €
1.2.35	1	PNEUMATIC PRESSURE BOOSTER	1.000,00 €
1.2.36	1	AIR DRYER FOR COMPRESSED AIR SUPPLY	5.800,00 €
1.2.37	1	SPECIAL MACHINE COLOUR	3.450,00 €

1.3 COMPONENT RELATED EQUIPMENT

1.3.2	1	SET OF TURNING TOOL HOLDER (SOFT) - BJ#22 & BJ#23 & CUJ#27	6.450,00 €
1.3.3	1	SET OF TURNING TOOLS (SOFT)	3.450,00 €
1.3.4	1	SET OF MILLING TOOLHOLDERS HSK 63-C - BJ#22	4.600,00 €
1.3.5	1	SET OF MILLING TOOLHOLDERS HSK 63-C - BJ#23	4.600,00 €
1.3.6	1	SET OF MILLING TOOLHOLDERS HSK 63-C - CUJ#27	4.600,00 €
1.3.7	1	SET OF MILLING TOOLS FOR ROUGH MILLING	12.300,00 €
1.3.8	1	SET OF MILLING TOOLS FOR FINISH MILLING	14.100,00 €
1.3.9	1	TEST BARS HSK63-C AND CAPTO C6	2.500,00 €
1.3.10	1	SETTING DIE FOR MILLING HEAD EXCHANGE	825,00 €
1.3.11	1	EQUIPMENT FOR BALL TRACK MILLING MACHINE	1.950,00 €
1.3.13	2	ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER SIN85 - with 1 medium duct	10.500,00 €
1.3.14	2	AIR SENSING	10.800,00 €
1.3.17	2	COLLET CHUCK (EMUGE) - ExaClamp	23.200,00 €
1.3.18	2	SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - BJ#22	10.000,00 €
1.3.19	2	SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - BJ#23	10.000,00 €
1.3.20	2	SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - CUJ#27	10.000,00 €

1.4 SERVICES

1.4.2	1	PROJECT ENGINEERING	37.100,00 €
1.4.3	1	TOOLING LAYOUT FOR OUTER RACE BJ#22 - SOFT	8.700,00 €
1.4.4	1	TOOLING LAYOUT FOR OUTER RACE BJ#23 - SOFT	8.700,00 €
1.4.5	1	TOOLING LAYOUT FOR OUTER RACE CUJ#27 - SOFT	8.700,00 €

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1.4.6	1	MACHINE SETUP FOR OUTER RACE BJ#22 - SOFT	12.500,00 €
1.4.7	1	MACHINE SETUP FOR OUTER RACE BJ#23 - SOFT	12.500,00 €
1.4.8	1	MACHINE SETUP FOR OUTER RACE CUJ#27 - SOFT	12.500,00 €
1.4.9	1	PRE-ACCEPTANCE OF THE MACHINE - SOFT MACHINING	9.200,00 €
1.4.10	1	PRE-ACCEPTANCE OF THE MACHINE - SOFT MACHINING	9.200,00 €
1.4.11	1	FUNCTION TEST OF THE MACHINE - SOFT MACHINING	3.350,00 €
1.4.12	1	INSTALLATION AND DEMONSTRATION (OVERSEAS)	35.000,00 €
1.4.13	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR BJ#22 - SOFT	5.500,00 €
1.4.14	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR BJ#23 - SOFT	5.500,00 €
1.4.15	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR CUJ#27 - SOFT	5.500,00 €
1.4.16	1	PRODUCTION SUPPORT AFTER START OF PRODUCTION	12.250,00 €
1.4.21	4	ADDITIONAL SET OF DOCUMENTATION	4.000,00 €
1.4.22	3	DOCUMENTATION ON USB	375,00 €
1.4.26	1	COMPLIANCE SPECIFICATION SHEET / DELIVERY CONDITIONS	11.300,00 €
1.4.28	1	OPERATING COURSE (Basic training)	6.125,00 €
1.4.29	1	MAINTENANCE (MECHANICAL AND ELECTRICAL) SERVICE COURSE (Basic training)	6.125,00 €
1.4.31	1	PACKING AND TRANSPORTATION OF TEST PIECES	4.800,00 €
1.4.32	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGES - KOR --> GER	10.200,00 €
1.4.33	1	PACKING OF INSPECTION GAUGES AFTER FINAL ACCEPTANCE	1.000,00 €
1.4.34	1	FOB HAMBURG	30.700,00 €
Total price:			1.413.225,00 €
Unique special discount:			-403.225,00€

1.5 MACHINE SPECIFICATIONS

Final price **1.010.000,00€**

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1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 DUO KBU

1.1 BASIC MACHINE

- 1.1.1 1 MINERALIT (CAST GRANITE) MACHINE BED
consisting of two separate sections, to prevent interaction during the machining process. For transport purposes the two sections are mechanically coupled.
The machine bed is manufactured from high quality Mineralit (cast granite) which guarantees superior thermal stability and excellent dampening characteristics. Mineralit has excellent vibration dampening properties which are six to eight times better than cast iron. The large distance between the two high precision pre-stressed linear roller guideways of the X-axis guarantees high accuracy and high dynamics. The guideways are outside the working area and therefore require no protective covers.
- 1.1.2 1 ENERGY CONTAINER WITH CONTROL CABINET
Electrical cabinet, cooled with water/air heat exchanger, equipped to VDE 0113 and EN 13849 and integrated in the over-all machine frame, thus ensuring a totally transportable unit. 400 V / 50 Hz machine voltage.
Other voltage on request.
The central grease supply system for lubrication of the recirculating ballscrews and guideway systems and also the service unit for compressed air supply are integrated in the energy container.
- 1.1.3 1 CNC-CONTROL SIEMENS
According to control specification.
- 1.1.4 1 HYDRAULIC UNIT
The hydraulic unit is equipped with a self regulating pump.
Supply 15 l/min (at 70 bar)
Tank capacity 63 l
- 1.1.5 1 COOLING UNIT (15 KW, water - air)
A cooling unit controls the thermal stability of the machine. Spindle bearings, drive unit and motor spindle for milling head are maintained at ambient temperature (10°C - 40°C).
Cooling capacity: 15 kW
- 1.1.6 1 MACHINE ENCLOSURE
The machining area is effectively separated from the load/unload area by the two side walls of the Mineralit machine base and a sliding door to the pick-up station.
An additional bellows completes the working area when loading and unloading the main spindle. Vapors generated during processing can't get into the environment.
A large door at the front of the machine provides access to the machining area (one for each spindle). The doors are equipped with a safety glass window to DIN EN ISO 23125:2010 specification.
Two inspection lamps are installed in each machining area (one at the B-axis and one at the spindle headstock cover traveling in X)

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- 1.1.7 2 **MULTI-FUNCTIONAL CROSS-SLIDE WITH Y-AXIS**
For the complete machining with driven tools or with drilling, milling, reaming or grinding spindles.
Large stroke of the Y-axis.
This unit is equipped with three axes (X, Y and Z) for each spindle. Both spindles can be programmed independently via the CNC control.
High rapid feed enable short idle times by component load/unload and gauging (optionally).
The axes drives are quick reaction frequency controlled service free servomotors which drive the slides over high precision ballscrew spindles. The X-, Y- and Z-axis are equipped with completely enclosed linear measuring systems.
On activation of the emergency-stop or in case of power failure, a brake will stop the vertical axis.
- 1.1.8 2 **MOTOR SPINDLE**
Spindle headstock unit with motor spindle
The motor spindle is a high dynamic frequency controlled service free AC synchronous spindle motor.
High degree of spindle rigidity due to triple bearing support over a short distance.
Precision shoulder bearings arranged in tandem (double-O). Additional bearing at spindle end for length compensation. All bearings are lifetime lubricated and thus maintenance free.
Sealing air to prevent dirt from penetrating from the outside.
The guideway system in the Z axis is equipped with hydrostatic bearings for compensation the cutting forces which guarantee excellent guideway quality by minimum friction.
Thermo-symmetrical design and integrated coolant system guarantee constantly maintained accuracy.
Positioning accuracy of spindle: +/- 0,01 angular degrees (C-axis)
Electro-hydraulic spindle locking optionally.
- 1.1.9 2 **MILLING HEAD FOR MILLING OF BALL GROOVES**
with 2 milling spindles
consisting of:
one axis, NC controlled swivelling unit (B-axis)
Swivelling housing guided by roller bearings in the side walls of the machine base.
Swivelling angle -30° to +90° (120°)
Swivelling drive by torque drive. Direct measuring system.
- 2 MOTOR SPINDLES HPC190 (manually tool clamping)
Max. power 40 % ED kW 15
Torque at 40 % ED Nm 67
Max. spindle speed rpm 8.500
Full power at spindle speed rpm 3.000
Spindle taper DIN 69893 HSK C63 / F80
Fluid-cooled spindle housing. Hybrid bearing.
Lifetime lubricated. Temperature controlled.
Rotating components are precision-balanced in two planes.
Coolant supply to tool holder pocket with minimal quantity lubrication max. 10 bar.
Pneumatic spindle nose cleaning device. Air seal between spindle and housing.
(warranty period 18 months)

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1.1.10	1	EMAG OPERATING AND SETTING MASKS The EMAG masks offer the setter/ operator optimal support in setting and operating the machine. Multiple base- and monitoring functions including data input are, depending on the machine type, available.	
1.1.11	1	For simple and fast programming of the machine on an appropriate ball track, the machine can be equipped with workshop-oriented programming modules. These modules are not included in the basic machine price and are described separately in the chapter „options to the CNC control“. On the other hand, the part programs can also be completed by the operator in normal ISO code.	
1.1.12	1	STANDARD COLOUR Machine - light grey to RAL 7035, front door - grey to RAL 7046, automation, swarf conveyor - grey to RAL 7046.	
1.1.13	1	NOISE LEVEL A maximum average corrected sound level of 78 dB(A) to DIN 45635 is obtained when turning with a continuous cut on a C45 steel disc in both longitudinal and transverse direction. If the workpiece has for example an interrupted cut due to its geometry or its clamping condition there may occur a higher sound level. For an extra cost sound deadening material can be fitted to reduce this higher sound level.	
1.1.14	1	PRICE OF MACHINE VSC 315 DUO KBU (SIEMENS) - SOFT MACHINING	734.700,00 €
1.2		ADDITIONAL EQUIPMENT	
1.2.1		TOOLING SYSTEM	
1.2.2	4	TOOL BLOCK FOR 1 TURNING TOOL CAPTO C6 manually clamped consisting of: toolholder with clamping insert C6 With internal air blast feeding to the tools. The coolant system guarantees constantly maintained accuracy.	61.600,00 €

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**1.2.3 COMPONENT STORAGE AND TRANSPORT SYSTEM**

- 1.2.4 1 RAW-PART STORAGE UNIT (SUPPLIED BY CUSTOMER)
Located to the left side of the machine.
Laid out for:
- approx. 340 workpieces of Outer Race BJ#22 (OD76.5mm)
- approx. 315 workpieces of Outer Race BJ#23 (OD80.4mm)
- approx. 207 workpieces of Outer Race CUJ#27 (OD99mm)
Chain-type conveyors working in opposite direction and transport the workpieces to the feed conveyor.
The feed conveyor is included.
- 1.2.5 1 LINEAR GANTRY LP3, WITH ROTATION MODULE 23.300,00 €
2-axis linear gantry
Drive by NC servomotor. Direct connection to the machine.
Energy guiding on all axes via cable track.
Rotation unit driven by a servo motor with rotary inlet.
Workpiece transport with a pneumatic 3-finger centric gripper
incl. 1 set of workpiece-specific gripper jaws.
without position detection and sensor gripper monitoring.
Prepared for fitting of an automatic alignment device for the correct angular positioning of workpieces in the ball tracks (hard machining).
- Swivel angle $\pm 360^\circ$
X-axis stroke 630 mm
Z-axis stroke 500 mm
Workpiece weight max. 3,5kg
- 1.2.6 1 GRIPPER JAWS OUTER RACE CLAMPING DIA. 20-36
1 set of gripper jaws for 3-finger centric part gripper
- Included in pos. "LINEAR GANTRY LP3, WITH ROTATION MODULE"**
- 1.2.7 1 POSITION DETECTION 2.550,00 €
Position detection of the workpiece via Keyence laser sensor, mounted on the gantry frame LP3
- 1.2.8 1 NIO - CHUTE, LENGTH 1550mm, R-L / L-R 3.300,00 €
Unloading, flow direction R-L / L-R
Design as a chute with structural sheet metal
For non-oriented or wrong workpieces, consisting of:
Direction to the side $\rightarrow$ right / left
1x light beam
Length: approx. 1550 mm
Accumulation range: approx. 1350 mm
- 1.2.9 1 SHUTTLE ZR120DUO L-R/ L-L FOR WORKPIECE LOADING 40.700,00 €
Type: ZR120DUO-A-1600-1305-800-LE-B-1
For the workpiece transport through the machine, with a basic pallet for blank parts.
Consists of a spring-loaded pick-up station, including detection and emergency stop.
Slip / brake plate coated with spring steel, 1 through for dry processing.
Mounting kit on the machine base body, 1x light barrier.
Driven via servomotor.
The pick-up spindle grasps the blank parts of the shuttle.
The workpieces are positioned NC-controlled for the pick-up spindle. Different workpiece diameters and heights are defined in the NC part program.

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1.2.10	2	WORKPIECE CARRIER EXPANDABLE OUTER RACE BJ#22 for the radial location of outer races by spring-actuated balls, for special applications. Designed for one workpiece.	7.900,00 €
1.2.11	2	WORKPIECE CARRIER EXPANDABLE OUTER RACE BJ#23 for the radial location of outer races by spring-actuated balls, for special applications. Designed for one workpiece.	7.900,00 €
1.2.12	2	WORKPIECE CARRIER EXPANDABLE OUTER RACE CUJ#27 for the radial location of outer races by spring-actuated balls, for special applications. Designed for one workpiece.	7.900,00 €
1.2.13	2	AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK For the cleaning of the pick-up unit and the clamping chuck by air blast. The air is processed by a service unit and directed at the component through nozzles.	2.400,00 €
1.2.14	1	WORKPIECE UNLOADING by motorised, chain-type conveyor Workpiece diameter: 35 - 120 mm Length: approx. 5000 mm Conveyors mounted to right of machine. Quick-change facility to adjust loading device to different workpiece diameters. 1x pneumatic stopper / separator ahead curve, adjustable. Light beams 1x workpiece stop. Electric / pneumatic installation on plug distributor.	28.500,00 €
1.2.15	1	LIGHT BEAM FOR UNLOADING CONVEYOR 1 x workpiece at the unloading position For external unloading	900,00 €

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VAT No.: DE 813 898 840

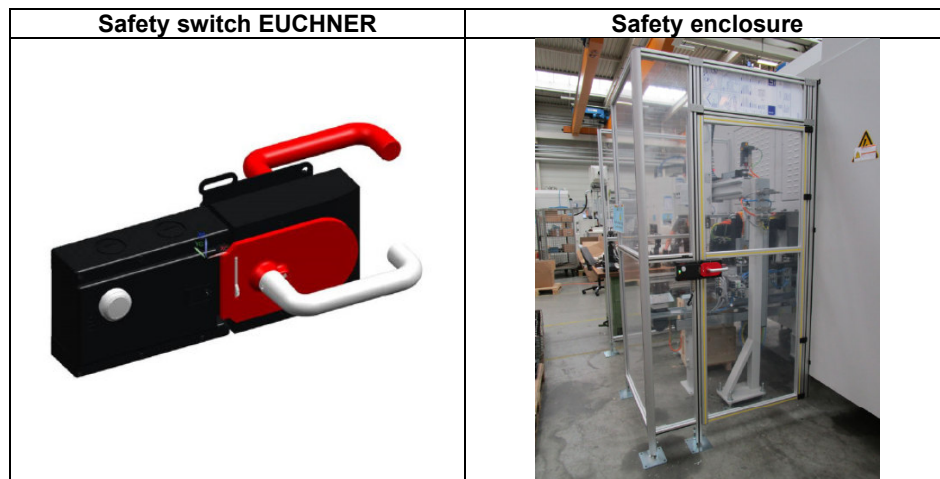
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1.2.16 1 SAFETY ENCLOSURE 12.100,00 €

Design consisting of:

- Aluminium profile (size 40)
 - Height of the safety fence: = 2400 [mm]
 - Space between ground and safety fence: 180 [mm]
 - Corner post quarter-round, all others square
 - Inserts Makrolon-discs, no woven wire mesh
 - Safety fence painted as follows:
 - Groove covers in door: yellow (RAL 1018)
 - Groove covers other fields: black
 - Makrolon discs: transparent 5mm thick
 - Visible profile ends covered with black plastic caps.
 - 4x Safety switch MGB Set EMAG, type MGB-L1HE-AR-R/L-EMAG, Maker: EUCHNER
- Enclosure covering moving automation components. Four access doors mechanically and electrically lockable.



Price does not include options (*)

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EMAG Maschinenfabrik GmbH

Austraße 24
73084 Salach - Germany
Phone: +49 7162 17-0
Fax: +49 7162 17-4270
E-Mail: info@emag.com
Internet: www.emag.com

EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

1.2.17 COOLANT AND SWarf DISPOSAL SYSTEM

- | | | | |
|--------|---|---|-------------|
| 1.2.18 | 1 | 2 SWarf CONVEYORS, CHAIN TYPE
Consisting of two single swarf conveyors:
The first swarf conveyor is mounted laterally on the unloading side of the machine.
Swarf exit height 350 mm
The second swarf conveyor carries the swarfs to the rear side of the machine.
Swarf exit height 1250 mm. | 27.400,00 € |
| 1.2.19 | 2 | CENTRAL EXTRACTOR UNIT (PREPARATION)
Connector for central extractor unit, mounted at the machining area.
Diameter 150 mm. Required extractor capacity approx. 1000 m ³ /h

Supply of a suitable extractor unit, for setup and pre acceptance of the machine at manufacturer's plant, by the customer or of EMAG at extra costs. | 1.650,00 € |
| 1.2.20 | 1 | CHIP BOX (LARGE VERSION)
with caster | 2.750,00 € |
| 1.2.21 | 1 | AIR GUN
with hose in blue color. | 1.050,00 € |



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**1.2.22 OPTIONAL EQUIPMENT FOR CNC CONTROL**

1.2.23	1	NC-GUIDED EMERGENCY RETRACTION AT POWER FAILURE (ESR) Interpolated retraction of axis in case of power failure, voltage fluctuation and drive failure. Electric components (condenser modul) are used which maintain the voltage for the duration of the retreat and use the energy for the nc-guided retract of the axes. (strongly recommended for unstable power supply, expensive tools or workpieces)	10.500,00 €
1.2.24	1	NETWORK DRIVE CONNECTION (Control system Siemens 840Dsl) Linking the machine control to your network after startup in your factory, so that network drives are available in the user interface. NC-programs or any data can be stored or loaded from there. Customer-specific requirements / restrictions for network integration: - Operation with DHCP server yes / no; if no, then IP address, sub-net mask and default gateway - DNS server and enable code of network source (also IP-address of server, if DNS not known) - User name and password for the automatic preparation of the connection - Network cable into the electrical cabinet When providing server and directory names it has to be taken into account that the 840Dsl features an English language Windows operating system that cannot handle the German umlauts ä, ö and ü. When installing of security update, the company Siemens restrictions are to consider. If in the domain automatically software is installed on all computers, so the functionality is no longer guaranteed after restart!	2.000,00 €
1.2.25	2	INTERFACE TO EXTERNAL HANDLING SYSTEM - HARDWARE Safety signal interface via HAN24E plug on electrical cabinet with 8 potential-free inputs and 8 potential-free outputs for the change of safety signals "emergency-stop" and "access door open" Handling-signal interface via HAN16E plug on electrical cabinet Deviations of the standard interface on request at additional costs. (1.000,00 € / signal)	7.400,00 €
1.2.26	1	MINI MANUAL CONTROL UNIT (SIEMENS) For operating a combined positioning and transfer unit and/or a linear gantry loader. In the setting up mode of the automation components: the shuttle, the vertical and horizontal axis of the gantry, also the swivelling and gripping function of the part gripper can be manually operated with the manual control unit. If the doors of the safety enclosure are open, the acknowledgment key must be pressed on the manual control unit.	4.500,00 €
1.2.27	1	CONNECTING KIT FOR MINI HAND-HELD CONTROL UNIT – SIEMENS	825,00 €
1.2.28	1	WORKSHOP-ORIENTATED PROGRAMMING AC-JOINTS - SOFT MACHINING For outer races (straight grooves). For quick and easy programming parameters for componentrelated dimensions are adapted. Same process for tool-offset. Every tool is allocated an empirically determined preset quantity of workpieces to be machined. On expiry of the tool life the machine is stopped. Control monitor display: tool-no., preset and remaining tool life quantity of workpieces per tool.	6.000,00 €

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1.2.29	1	WORKSHOP-ORIENTATED PROGRAMMING CUJ-JOINTS - SOFT MACHINING For outer races (straight grooves). For quick and easy programming parameters for componentrelated dimensions are adapted. Same process for tool-offset. Every tool is allocated an empirically determined preset quantity of workpieces to be machined. On expiry of the tool life the machine is stopped. Control monitor display: tool-no., preset and remaining tool life quantity of workpieces per tool.	6.000,00 €
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1.2.30 COUNTRY SPECIFIC EQUIPMENT

1.2.31	1	SERIES TRANSFORMER (63A) (low version) for 3-phase power supply, with neutral conductor and ground conductor. Power supply (prim): 380, 420, 440, 460, 480 V - 50 / 60 Hz Other supply on request. Transformer in separate housing. Located on the machine. Customer pre-fuse required. Connection and link wiring provided by customer.	3.750,00 €
1.2.32	1	AUXILIARY COOLING SYSTEM consisting of auxiliary coolant supply unit for switchgear cabinet.	3.500,00 €

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**1.2.33 OTHER AUXILIARY EQUIPMENT**

1.2.34	1	4-COLOUR STATUS LIGHT (Q-LIGHT) mounted on machine. Colour code: red: machine stopped yellow: warning -tool to be changed, -lubricant running low, etc. green: machine in automatic cycle. white: M/C ready (waiting for workpiece supply)	1.200,00 €
1.2.35	1	PNEUMATIC PRESSURE BOOSTER to increase air pressure from 4 bar to 6 bar. (only for part grippers) Maker: Festo Type: DPA-63-10	1.000,00 €
1.2.36	1	AIR DRYER FOR COMPRESSED AIR SUPPLY Absorption air dryer for dry and clean air. Manufacturer: Festo Type: PDAD-51-G1/2 incl. combined maintenance unit	5.800,00 €
1.2.37	1	SPECIAL MACHINE COLOUR Complete machine according to customers colour specification.	3.450,00 €

1.3 COMPONENT RELATED EQUIPMENT**1.3.1 TOOL EQUIPMENT**

1.3.2	1	SET OF TURNING TOOL HOLDER (SOFT) - BJ#22 & BJ#23 & CUJ#27 consisting of: - 2 turning tools Capto C6 - rough turning - 2 turning tools Capto C6 - finish turning - 2 turning tools Capto C6 - rough turning - as spare - 2 turning tools Capto C6 - finish turning - as spare	6.450,00 €
1.3.3	1	SET OF ROUGH TURNING TOOLS (SOFT) consisting of: - 40 carbide cutting inserts - rough turning - 40 carbide cutting inserts - finish turning	3.450,00 €
1.3.4	1	SET OF MILLING TOOLHOLDERS HSK 63-C - BJ#22 Toolholders for interchangeable milling heads. - 2 pcs rough milling holder for BJ#22 (2x on the MC) - 2 pcs finish milling holder for BJ#22 (2x on the MC) - 2 pcs rough milling holder for BJ#22 (2x as spare) - 2 pcs finish milling holder for BJ#22 (2x as spare)	4.600,00 €
1.3.5	1	SET OF MILLING TOOLHOLDERS HSK 63-C - BJ#23 Toolholders for interchangeable milling heads. - 2 pcs rough milling holder for BJ#23 (2x on the MC) - 2 pcs finish milling holder for BJ#23 (2x on the MC) - 2 pcs rough milling holder for BJ#23 (2x as spare) - 2 pcs finish milling holder for BJ#23 (2x as spare)	4.600,00 €

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1.3.6	1	SET OF MILLING TOOLHOLDERS HSK 63-C - CUJ#27 Toolholders for interchangeable milling heads. - 2 pcs rough milling holder for CUJ#27 (2x on the MC) - 2 pcs finish milling holder for CUJ#27 (2x on the MC) - 2 pcs rough milling holder for CUJ#27 (2x as spare) - 2 pcs finish milling holder for CUJ#27 (2x as spare)	4.600,00 €
1.3.7	1	SET OF MILLING TOOLS FOR ROUGH MILLING Head system with carbid cutting inserts - 30 cutting heads Ø21[mm] for BJ#22 - 30 cutting heads Ø25[mm] for BJ#23 and CUJ#27	12.300,00 €
1.3.8	1	SET OF MILLING TOOLS FOR FINISH MILLING Head system with carbid cutting inserts - 30 cutting heads Ø 16.0[mm] for BJ#22 - 30 cutting heads Ø 16.8[mm] for BJ#23 - 30 cutting heads Ø 20.3[mm] for CUJ#27	14.100,00 €
1.3.9	1	TEST BARS HSK63-C AND CAPTO C6 for checking the machine geometry Consisting of: 1x aluminum case 1x foam insert 1x test bar HSK63-C 1x test bar Capto C6	2.500,00 €
1.3.10	1	SETTING DIE FOR MILLING HEAD EXCHANGE consisting of : 1x fixture for milling head change for mounting on a bench.	825,00 €
1.3.11	1	EQUIPMENT FOR BALL TRACK MILLING MACHINE consisting of: - 1 cone wiper - 1 torque wrench with insert - 2 plugs HSK63 - 1 ball plug gauge HSK63	1.950,00 €

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**1.3.12 CHUCKING SYSTEM**

1.3.13 2 ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER SIN85 10.500,00 €

- with 1 medium duct

Max. drawbar force 3.325 daN at 70 bar.

The clamping pressure is manually infinitely adjustable. .

The clamping cylinder stroke and the clamping pressure are controlled by a linear measuring system and by a pressure switch.

1.3.14 2 AIR SENSING 10.800,00 €

The air sensing is exclusively used to identify the workpiece contact on the clamping device.

This is done by measuring the dynamic pressure that occurs between the workpiece and the clamping device.

The air is supplied by one or more nozzles (depending on clamping fixture type).

The pressure is measured with a pressure switch.

The air supply is controlled by a directional valve.

The air flow is sent through the main spindle.

NOTE:

Requires CHUCKING CYLINDER WITH MEDIUM DUCT

Equipping the clamping fixture with a pneumatic system monitor is not included in the price of this item.

1.3.15 COLLET CHUCK - DIRECT CONNECTION**COLLET CHUCK EMUGE ExaClamp**

1.3.16 1 COLLET CHUCK EMUGE ExaClamp

Please note that the following EMUGE collet chuck is a new standard chuck.

The clamping heads / collets from the older version are compatibel with the new standard chuck.

The part stops / datum plates and dirt covers are not compatibel.

1.3.17 2 COLLET CHUCK (EMUGE) - ExaClamp 23.200,00 €

For clamping of Outer Races and Outer Race Discs.

Installation direct on the main spindle (no spindle adapter required).

With basic part stop.

Clamping range: 65 mm - 120 mm

Prepared for air sensing.

Without workpiece-specifically interchangeable parts.

1.3.18 2 SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - BJ#22 10.000,00 €

Laid out for one Outer Race.

Consisting of clamping head, part stop, dirt cover and in total one control ring.

1.3.19 2 SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - BJ#23 10.000,00 €

Laid out for one Outer Race.

Consisting of clamping head, part stop, dirt cover and in total one control ring.

1.3.20 2 SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - CUJ#27 10.000,00 €

Laid out for one Outer Race.

Consisting of clamping head, part stop, dirt cover and in total one control ring.

1.4 SERVICES

Price does not include options (*)

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1.4.1	1	Please note that no additional services are contained in the previously specified prices. We would be pleased to carry out the following services and would quote according to your wishes. - Preparation of tooling layouts and NC programs - Set-up machine for component trials on your specific workpiece in our works - Machine approval in our works by your engineers. - Final acceptance at customer's plant with a specific workpiece	
1.4.2	1	PROJECT ENGINEERING Project engineering and coordination with suppliers and customers. Engineering investigations (machine set up, with central automation, central exhaust, chip disposal) Customization of interfaces with the suppliers.	37.100,00 €
1.4.3	1	TOOLING LAYOUT FOR OUTER RACE BJ#22 - SOFT Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.	8.700,00 €
1.4.4	1	TOOLING LAYOUT FOR OUTER RACE BJ#23 - SOFT Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.	8.700,00 €
1.4.5	1	TOOLING LAYOUT FOR OUTER RACE CUJ#27 - SOFT Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.	8.700,00 €
1.4.6	1	MACHINE SETUP FOR OUTER RACE BJ#22 - SOFT Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	12.500,00 €
1.4.7	1	MACHINE SETUP FOR OUTER RACE BJ#23 - SOFT Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	12.500,00 €
1.4.8	1	MACHINE SETUP FOR OUTER RACE CUJ#27 - SOFT Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	12.500,00 €
1.4.9	1	PRE-ACCEPTANCE OF THE MACHINE - SOFT MACHINING Specific workpiece: Outer Race BJ#22 with measuring equipment supplied by customer. One workpiece in one clamping operation. Max. 30 components. Submission of proof to Cm of machine's capability of max. 3 measuring characteristics. For Cm of machine's capability measurements of workpiece 1, 5, 10, 15, 20, 25, 30.	9.200,00 €
1.4.10	1	PRE-ACCEPTANCE OF THE MACHINE - SOFT MACHINING Specific workpiece: Outer Race CUJ#27 with measuring equipment supplied by customer. One workpiece in one clamping operation. Max. 30 components. Submission of proof to Cm of machine's capability of max. 3 measuring characteristics. For Cm of machine's capability measurements of workpiece 1, 5, 10, 15, 20, 25, 30.	9.200,00 €

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1.4.11	1	FUNCTION TEST OF THE MACHINE - SOFT MACHINING Specific workpiece: Outer Race BJ#23 for one workpiece including test run, without SPC analysis, 10 parts machined in 100% drawing tolerance. With measuring equipment supplied by customer.	3.350,00 €
1.4.12	1	INSTALLATION AND DEMONSTRATION (OVERSEAS) in your works The following services are included: - Machine installation and demonstration - Machine approval and hand-over - Talk-through of the machine documentation and service instruction - Operator instruction - Travel costs Total of 15 man days Not included are: - Machine set-up - Machine operation for production - Additional service Additional service engineer costs for time in excess of that quoted will be charged extra.	35.000,00 €
1.4.13	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR BJ#22 - SOFT Specific workpiece: Outer Race BJ#22 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces Evaluation and guaranteed criteria are the same as for pre-acceptance test.	5.500,00 €
1.4.14	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR BJ#23 - SOFT Specific workpiece: Outer Race BJ#23 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces Evaluation and guaranteed criteria are the same as for pre-acceptance test.	5.500,00 €
1.4.15	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR CUJ#27 - SOFT Specific workpiece: Outer Race CUJ#27 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces Evaluation and guaranteed criteria are the same as for pre-acceptance test.	5.500,00 €
1.4.16	1	PRODUCTION SUPPORT AFTER START OF PRODUCTION 1 week (6 days - monday - saturday) production support at customers plant. Performed by EMAG Mexico.	12.250,00 €
1.4.17	1	INSTALLATION POINT The installation point has to be prepared in accordance with the documentation supplied. We take it for granted that all conditions are met before our installation crew arrives. Where waiting times are incurred due to the conditions not having been met, these will be charged extra. During installation and commissioning our engineers have to be provided, on request, with the support of skilled customer personnel and similar assistance. This also applies to the operation of cranes, handling equipment and other aids.	

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- 1.4.18 1 **MACHINE LIFTING EQUIPMENT**
EMAG will provide a lifting equipment to lift the machine by crane, free of charge.
However, should the customer not return the equipment to EMAG within 4 weeks, freight
and charges prepaid, a separate invoice of EUR 1.500,00 will be issued.

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1.4.19 DOCUMENTATION

1.4.20 1 DOCUMENTATION (Siemens)

OPERATING INSTRUCTIONS

1 complete set of operating instructions on paper

EU countries in the official language of the country; other countries on national regulations

Manuals for the machine:

- Introduction
- Safety instructions
- Product description
- Transport and installation
- Start-up
- Production
- Preventive maintenance

Manuals from the manufacturer of the control unit:

Siemens SIN 840D sl

- User manual
- Diagnostics manual
- Programming - Basics
- Programming - Cycles
- Programming - Operations planning and scheduling

Machine diagrams:

- Machine layout
- Transport diagram
- Machine data sheet
- Drive data sheet
- Connection data sheet for fluidics
- Connection data sheet for electrics

List of spare parts and wearing parts

Operating instructions from third parties

Documentation for clamping devices and tools

TECHNICAL DATA

1 set of technical data on paper, German or English

- Electrical circuit diagram with list of el. devices
- Fluid system diagram with list of el. devices
- Drawings of subassemblies with relevance for spare parts
- List of components with relevance for spare parts and wearing parts in relation to the drawings
- Technical data, test reports, declarations of incorporation from third parties

Other languages available on request for an additional charge; third party documentation excepted

1.4.21	4	ADDITIONAL SET OF DOCUMENTATION 1 additional set of the documentation described above. Language: 3x Spanish, 1x Korean	4.000,00 €
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Price does not include options (*)

Offer : 196563 / 5

Managing Directors:
Dr. Guido Hegener, Dr. Mathias Klein
Commercial Register Ulm: HRB 533796
Tax No.: 63002/12866 FA Göppingen
VAT No.: DE 813 898 840

HYUNDAI WIA Corporation

Bank accounts:
UniCredit Bank AG / Hypovereinsbank Stuttgart 'IBAN DE02 6002 0290 0322 6388 17' BIC HYVEDEMM473
Commerzbank AG Tübingen 'IBAN DE09 6414 0036 0890 9103 00' BIC COBADEFF641
Kreissparkasse Göppingen 'IBAN DE96 6105 0000 0015 5885 93' BIC GOPSDE6GXXX
Bayerische Landesbank München 'IBAN DE59 7005 0000 0004 2569 67' BIC BYLADEMMXXX

19/08/2022

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EMAG Maschinenfabrik GmbH

Austraße 24
73084 Salach - Germany
Phone: +49 7162 17-0
Fax: +49 7162 17-4270
E-Mail: info@emag.com
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EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

1.4.22	3	DOCUMENTATION ON USB 1 complete set of documentation on USB Language: 2x English, 1x Spanish	375,00 €
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1.4.23 CE-MARKING

- 1.4.24 1 WITHOUT CE MARKING / WITHOUT DECLARATION OF CONFORMITY (non-EU country, complete machine)
The machinery is always designed and constructed in accordance with the applicable harmonized standards and safety instructions. For operation of the machine, the regional regulations have to be observed additionally.

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EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

1.4.25

CUSTOMER SPECIFICATION

1.4.26

1

COMPLIANCE SPECIFICATION SHEET / DELIVERY CONDITIONS

Design of mechanical-, electrical- and fluid equipment according to "M/C Specification, Date: 1st July 2022 - WGT-M4-0022-001 - 1st version"

Due to our design we are unable to comply with the following items of your specification sheet, meaning we must apply for an exceptional permission.

11.300,00 €

Item	Customer requirement	EMAG commitment/Exceptional approval
Basic Condition	Surrounding temperature: 0°C - 40°C	Surrounding temperature: +10°C to +40°C
Noise level	less than 75 dB (A)	up to 78dB (A)
General specification of Hyundai WIA	Keep the "General specification of Hyundai WIA"	This document is not part of the inquiry and therefore it is not considered.
Main Circuit	Solenoid valve: 110V AC 60Hz, Lamp Type, Communication Type	24V DC with LED
CNC & PLC	- Voltage 240V - Memory USB Card - Memory EPROM	400V AC / 24V DC - Memory USB-Stick - No longer available from Siemens.
CNC & PLC	Network interface - LAN	USB
Wiring	Remain the 30% marginal space and do not wire it hose together	max. 10% possible
Items for the capability (pre- and final acceptance)	Ball Track Milling: Groove Dia, Center, Ball Diameter	Ball diameter is no item for evaluation (Cm)
Detail 14. Others (Lub) &	2) Hyd. Pump - The type of hyd. Pump should be variable discharge 4) Inlet The mesh could be #100-120 mesh 2) Maker should install the air dryer to remove the water. Specification of air dryer: SMC-IDF15E1 14) The chuck maker should be EMUGE (for common use for existing line) The pallet for raw part should be pin type. Neither ball nor chuck type is allowed. 33) LMS should be established and connect to MES. "Work name, Status (machining, waiting, Stop), counter, feedrate, C/Time etc. 36) When SPC time comes, m/c alarm after discharge of the workpiece, and then starts automatically after SPC is done.	2) Pump engine is frequency controlled. 4) EMAG use 10µm 2) EMAG use FESTO membrane air dryer MS6-LDM1-1/2-P20. Please either grant us an exemption or provide us with more details about your requested air dryer SMC-IDF15E1 14) EMUGE has a new standard collet chuck ExaClamp. Only the clamping collets are compatible. The datum's/ part stops and dirt covers are not compatible. The pallet for raw part (soft machining) are spring actuated ball clamps. Same design as delivered with MC SN-108818. The tolerance of the preforged ball tracks is too large for rigid workpiece carriers  33) Same functionality and scope as in the delivered machines P-30112294/SN108817 and P-30112295/SN108818. No individual tracking of workpieces considered. 36) The operator must manually confirm after removing and checking the SPC workpieces.
Safety Fence	According to WIA Specification.	Please refer to pos. 1.2.16 in this offer. For some items an exception approval is required.
Electric	OP Panel: Height = FL + 1300mm	Height of the OP Panel: Keys approx. 1200mm to 1440mm Screen approx. 1450mm to 1750mm

Price does not include options (*)

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EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

1.4.27 TRAINING

- | | | | |
|--------|---|--|------------|
| 1.4.28 | 1 | OPERATING COURSE (Basic training)
For two participants in your works, performed by EMAG Mexico.
Matched to the machine design (modular customized)
Training course duration: 3 days / shift (shift monday - saturday).
Language: Spanish
Documentation: 4x Spanish | 6.125,00 € |
| 1.4.29 | 1 | MAINTENANCE (MECHANICAL AND ELECTRICAL) SERVICE COURSE (Basic training)
For two participants in your works, performed by EMAG Mexico.
Matched to the machine design (modular customized)
Training course duration: 3 days / shift (shift monday - friday).
Language: Spanish
Documentation: 4x Spanish | 6.125,00 € |

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1.4.30 TRANSPORT COSTS

1.4.31	1	PACKING AND TRANSPORTATION OF TEST PIECES Transport (ocean freight) from customers plant in Mexico or Korea to EMAG Maschinenfabrik in Salach. - 300 test pieces of acceptance part - 150 test pieces for the 1st function test part - 150 test pieces for the 2nd function test part. Max. total weight = 950 [kg]. The customer has to take care of a transport-safe packaging in each case.	4.800,00 €
1.4.32	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGES - KOR --> GER Transport from customers plant in Korea to EMAG Salach Germany. (Air freight, max. 750kg) The customer has to take care of a transport-safe packaging in each case.	10.200,00 €
1.4.33	1	PACKING OF INSPECTION GAUGES AFTER FINAL ACCEPTANCE Packing of the inspection gauges after final acceptance by EMAG employees. Inspection gauges will be send together with the machine to customers plant. Responsible for seaworthy packing is the customer Hyundai WIA. It must be ensured that EMAG can reuse the seaworthy packaging.	1.000,00 €
1.4.34	1	FOB HAMBURG from Salach, Incoterms 2020	30.700,00 €
Total price:			1.413.225,00 €
1.4.35	1	UNIQUE SPECIAL DISCOUNT	-403.225,00 €

1.5 MACHINE SPECIFICATIONS

Price does not include options (*)

Offer : 196563 / 5

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1.5.1 1 TECHNICAL MACHINE DATA VSC 315 DUO KBU Subject to change without notice

WORKING AREA

Chuck diameter	mm	250
Swing diameter	mm	260
X-stroke	mm	900
Y-stroke	mm	315
Z-stroke	mm	300

WORKPIECE DIMENSIONS

Workpiece length, max.	mm	260*
Workpiece O/D, max.	mm	120
Ball section dia., max.	mm	30

(*Longer workpieces on request)

MAIN SPINDLE

Spindle flange acc. to DIN 55026	size	8
Spindle front bearing diameter	mm	140
max. spindle speed	rpm	3000

MAIN DRIVE

AC-Synchronous motor 40/100 % ED	kW	28,5/27,5
Max. power	kW	28,5
Full power at spindle speed	rpm	700
Torque at 40/100 % ED	Nm	590/375
Max. torque at	Nm	590

FEED DRIVE

Rapid traverse feed, X+Y	m/min	45
Rapid traverse feed, Z	m/min	30
Feed force, X/Y/Z	kN	11
Ball screw diameter X/Y/Z	mm	50/40/40

MILLING SPINDLE

AC-motor	kW	15
Max. spindle speed	rpm	8500

ELECTRICAL EQUIPMENT according to VDE 0113

Supply voltage	V	400
Control circuit voltage - DC	V	24
- AC	V	230
Frequency	Hz	50
Total connected load	kVA	120
Input fuse	A	250

DIMENSIONS AND WEIGHT

Length	mm	4400
Width	mm	3600
Height	mm	3500
Weight	kg approx.	20000

1.5.2 1 SIEMENS OPERATOR PANEL 15" MULTITOUCH MONITOR
consisting of OP015 Black + IPC 427E + Control panel (MCP483C)

Price does not include options (*)

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- 1.5.3 1 **CNC CONTROL SIEMENS**
SINUMERIK 840 Dsl with integrated PLC S7-300
Operator software Sinumerik Operate with HMI-Pro (Windows 10)
Screen language and on-screen messages in German / English / in official EU language,
more languages available as an optional
CNC program and data memory 10 Mbyte
USB-interface (front) for memory-stick (USB 2.0)
Machine diagnostics via internet with VPN technology to verify EMAG's standard format
Software backup is to external media (USB stick integrated into IPC) to verify EMAG's
standard format
dataGrabber (Comara) – Capable of recording process data for diagnostics, optional
process optimization
No active evaluation possible on the control, storage of the data takes place in a ring
buffer

EMAG ADD-ON FUNCTIONS:

Machine-specific documentation of functions and parameters is available directly on the
control unit. (PDF display)
Automatic return to the home position upon machine error. Error recovery is a simple one
button process.
First run high speed traverse is at a reduced speed from home to the first cut. Reduced
speed allows you to avoid a collision and save damaged equipment and reprogramming
costs.
Internal measuring probe with collision monitoring on the traversing movements (Option)
Automatic workpiece gaging when switching to a replacement tool or tool insert (Option)
Automatic filling and emptying of the entire machine.
"Optional Stop" (M01) allows you to make a programmed stop or interrupt production after
finishing a part at a desired point.
Single cycle: Allows one part to be transferred and machined for setup and tryout
purposes.

STRUCTURED WORKPIECE PROGRAMMING:

Division of individual programs into subprograms, e.g. the program for the machining
contains only the technology program and not the loading part.
The loading program has individual parameters. Division into machine and work piece
specific parameters.
Programmable chuck with length measuring system in the pull rod.
Different clamping situations (Unclamped, Clamped and Over stroke) can be changed by
parameter assignment. Displayed by diagnostics menu.
Recording and displaying of the most important production data: work piece counter, shift
counter, cycle time, work piece status, operating hours.
Production of definable batch sizes. The machine stops upon reaching the selected no.
Optimized programming for the turret keeps chips from collecting.
Rinsing of clamping devices after a programmable number of parts
Number and positions of the workpiece carriers can be defined for closed automation.
Lifecycle screen that displays tool wear and tool life.

Final price:

1.010.000,00 €

Price does not include options (*)

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Commercial Conditions

CONFIDENTIALITY

Any commercial and operational written or oral information or data relating to this quotation / order, such as drawings, specifications, recommended machining data and clamping methods, etc., shall not be divulged to any third party (referring, in particular, but not exclusively, to competitors, for the purpose of bid comparisons) and are to be used solely within the meaning of this quotation / order.

The recipient of this quotation / order expressly accepts these obligations.

MACHINING

The determined cycle times result from values, which are based on data and experiences of acknowledged cutting material manufacturers. Changing rough part dimensions, tolerances, material as well as the material hardness are only possible during building and testing of the machine. Changes will only be accepted in written form and in mutual agreement.

PROVISION OF EQUIPMENT AND MATERIALS BY CUSTOMER

All tooling, work holding equipment, parts, etc. are to be supplied with DDP supplying plant EMAG Incoterms 2020. We wish to point out that we accept no responsibility for the functional capability and working order of any equipment provided.

The functional capability of the provided equipment is the responsibility of the buyer.

In the event of problems, we assure you of our constructive collaboration. Any additional costs arising from modification, optimization, etc. will be charged to the customer. Furthermore, we do not guarantee any quoted machining times in such cases.

When orders cannot be carried out on schedule, caused by material supplied by the customer or if technical problems should occur, the machine will be invoiced on the originally agreed delivery date, when the relevant payment will become due.

The agreed upon date of delivery date may be postponed to the time to achieve full operability, which shall not result in any event of default of the manufacturer.

DELIVERY PERIOD

Approx. 12 months FOB German Seaport when order will be placed until calendar week 34/2022. The delivery period is not binding and can change between the submission of quotation and placement of order.

Please discuss the subject of delivery with us at time of order placement and we will do everything possible to meet your requirements. Changes regarding the component/s or the machines or the order content after order placement may result in changes in price and delivery.

Compliance of delivery date requires timely down and / or partial payment by the customer within the prescribed payment terms. If receipt of down and or partial payment is delayed, the delivery shall be postponed accordingly, which shall not result in any event of default of the manufacturer.

CONFORMITY WITH CUSTOMERS` REGULATIONS

has not been taken into account. Where unavoidable deviations from such regulations prove necessary, specific authorization will be requested.

PRICES

FCA (Free Carrier, Incoterms 2020). Packing, insurance and transport not included.

The payments are to be made without deduction. With delayed payment the relevant laws apply.

VALIDITY OF PRICES

The prices quoted are valid for 90 days from the date of quotation. Orders placed after this period would be subject to a price increase of 0.5 % per month to cover increases in labour and material costs.

This quotation will replace all previous versions.

Price does not include options (*)

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TERMS OF PAYMENT

100 % of invoice value against irrevocable Letter of Credit payable as follows:
30 % of L/C-value after opening of the L/C as advance payment against simple receipt.
70 % of L/C-value at sight against shipping documents.
All L/C-charges are for buyer's account. L/C must be opened latest 30 days after order confirmation

RETENTION OF TITLE

The machine will remain the property of the builder until complete payment is received from the buyer. Upon placement of order the extended retention of title according to clause 48 of our General Conditions for Sales and Deliveries shall be deemed as agreed.

LIMITATION OF LIABILITY

The liability of the supplier is excluded for indirect damage, defects and consequential damage, or losses, e.g., loss of revenues, downtimes, production stoppages, capital costs, or costs incurred due to operational interruptions. This does not apply to damage willfully caused by the supplier, or in cases of statutory liability, in particular according to the Product Liability Act.

PERIOD OF LIABILITY

Any rights and claims of the buyer caused on defects shall be subject to a limitation period of twelve months. The limitation period starts with the final acceptance, but no later than 3 months from delivery, and does not apply to wear and tear parts.

WORK PIECES FOR MACHINE ACCEPTANCE

Both, pre-acceptance and final acceptance tests are carried out with either an EMAG test component to VDI standard, or the work piece specified by the customer's order.

The customer will supply an agreed number of work pieces for machining process optimisation and pre-acceptance tests. These work pieces must be from a single batch and of a quality that corresponds with the relevant manufacturing drawing. For preparatory work and preliminary machine acceptance the customer will, at the agreed time, supply EMAG free of charge with a sufficient number of sample work pieces of a quality that corresponds with the relevant manufacturing drawing. Any delay in the supply of these work pieces will result in the machine delivery being delayed and the starting date of a possible delayed machine delivery penalty being deferred accordingly.

FINAL ACCEPTANCE AND FINAL PAYMENT

Final acceptance tests are carried out as soon as the machine has been installed and commissioned. For this purpose, the buyer has to provide a sufficient number of work pieces, personnel and capacity to measure and evaluate the acceptance components. Once the machine is taken into production and the agreed component quality and cycle time have been achieved, final acceptance is considered to have taken place. Where final acceptance of the machine is not possible owing to reasons beyond EMAG's control (missing acceptance components, etc.), final contractual payment falls due immediately.

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TECHNICAL AVAILABILITY

Our confirmation of the technical availability according to VDI 3423 is based on following conditions:
Exclusively qualified operating personnel, monthly availability protocols addressed to our service manager, demonstrable preventive maintenance is carried out according to our machine documentation, connection to our tele service system if requested by our service department.
In case of non-compliance of these conditions, our commitment expires.

TECHNICAL DEVELOPMENT

In the interest of technical progress, we reserve ourselves the right to make design changes which are due to continuous product development.

RESERVATION CLAUSE

For offers within the European Union (without Germany):

„Notification of Intra-Community delivery according to Art. 22 (10) EC Dual-Use Regulation:

The following purchased goods are subject to export controls. For the export of these goods from the Community an export license is necessary.”

For offers outside the European Union:

It is agreed, that the legally binding conclusion of this offer is subject to the suspensive condition that the competent authority grant the necessary export license.

ADDITIONAL PROVISIONS

In addition and subordinated to this quotation our General Conditions for Sales and Deliveries for International Business shall apply (download under http://www.emag.com/Delivery_conditions_UN.pdf).

This document was created electronically and is valid without signature.

Sincerely,

EMAG Maschinenfabrik GmbH

Hartmut von Fugler
Managing Director

Do-Youn Koo
Area Sales Manager

Enclosures

EMAG-General conditions for sales and deliveries (UN)
Layout
Schematic of machining, TImestudy, Quality commitments
Part MAKER List Hyundai WIA Mexico_ Commented by EMAG
STN KV EI + Mech + Fluid HYUNDAI WIA_DG_V ; EMAG-Interface to external handling systems

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EMAG Maschinenfabrik GmbH

Austraße 24
73084 Salach - Germany
Phone: +49 7162 17-0
Fax: +49 7162 17-4270
E-Mail: info@emag.com
Internet: www.emag.com

EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

HYUNDAI WIA Corporation
Mr. Kyung Yang Lee
153, Jeongdong-ro,
Seongsan-gu,
Changwon-si,
Gyeongsangnam-do 51533
SOUTH KOREA

Offer

Offers- No: 196564 / 5
date: 19/08/2022

Your technology contact

Andreas Liebrich

Phone +49 7162 17-4693
E-mail aliebrich@emag.com

Your contact

Do-Youn Koo

Phone +82 31 766-4415
E-mail dykoo@emag.com

Your Reference

Inquiry dated 29/11/2021
Contact Kyung Yang Lee
Phone +82 55 2809501
E-mail ardanji@hyundai-wia.com

EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 DUO KBU for straight ball grooves (AC- and CUJ-joints) for the simultaneous hard turning and milling and two Outer Races of the same type

Outer Race BJ#22
Outer Race BJ#23
Outer Race CUJ#27



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Price does not include options (*)

Offer : 196564 / 5

Managing Directors:
Dr. Guido Hegener, Dr. Mathias Klein
Commercial Register Ulm: HRB 533796
Tax No.: 63002/12866 FA Göppingen
VAT No.: DE 813 898 840

HYUNDAI WIA Corporation

Bank accounts:
UniCredit Bank AG / Hypovereinsbank Stuttgart 'IBAN DE02 6002 0290 0322 6388 17' BIC HYVEDEMM473
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Price Summary

1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 DUO KBU

1.1 BASIC MACHINE

1.1.14 1 PRICE OF MACHINE VSC 315 DUO KBU (SIEMENS) - HARD MACHINING 654.700,00 €

1.2 ADDITIONAL EQUIPMENT

1.2.2	2	TOOL BLOCK FOR 1 TURNING TOOL CAPTO C6 manually clamped	30.800,00 €
1.2.3	2	PROTECTIVE PLATE FOR THE EMPTY TURNING TOOL POSITION	250,00 €
1.2.5	1	WORKPIECE LOADING by motorised, chain-type conveyor	22.200,00 €
1.2.6	1	WORKPIECE STOP OUTER RACE / DISC TYPE	0,00 €
1.2.7	1	LIGHT BEAMS FOR LOADING CONVEYOR	950,00 €
1.2.8	1	LINEAR GANTRY LP3, WITH ROTATION MODULE	23.300,00 €
1.2.9	1	GRIPPER JAWS OUTER RACE CLAMPING DIA. 20-36	0,00 €
1.2.10	1	POSITION DETECTION	2.550,00 €
1.2.11	1	NIO - CHUTE, LENGTH 1550mm, R-L / L-R	3.300,00 €
1.2.12	1	SHUTTLE ZR120DUO L-R/ L-L FOR WORKPIECE LOADING	40.700,00 €
1.2.13	2	WORKPIECE CARRIER RIGID OUTER RACE BJ#22	3.700,00 €
1.2.14	2	WORKPIECE CARRIER RIGID OUTER RACE BJ#23	3.700,00 €
1.2.15	2	WORKPIECE CARRIER RIGID OUTER RACE CUJ#27	3.700,00 €
1.2.16	2	AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK	2.400,00 €
1.2.17	1	WORKPIECE UNLOADING by motorised, chain-type conveyor	28.500,00 €
1.2.18	1	LIGHT BEAM FOR UNLOADING CONVEYOR	900,00 €
1.2.19	1	SAFETY ENCLOSURE	10.900,00 €
1.2.21	1	2 SWarf CONVEYORS, CHAIN TYPE	27.400,00 €
1.2.22	2	CENTRAL EXTRACTOR UNIT (PREPARATION)	1.650,00 €
1.2.23	1	CHIP BOX (LARGE VERSION)	2.750,00 €
1.2.24	1	AIR GUN	1.050,00 €
1.2.26	1	NC-GUIDED EMERGENCY RETRACTION AT POWER FAILURE (ESR)	10.500,00 €
1.2.27	1	NETWORK DRIVE CONNECTION	2.000,00 €

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1.2.28	2	INTERFACE TO EXTERNAL HANDLING SYSTEM - HARDWARE	7.400,00 €
1.2.29	1	CONNECTING KIT FOR MINI HAND-HELD CONTROL UNIT – SIEMENS	825,00 €
1.2.30	1	WORKSHOP-ORIENTATED PROGRAMMING AC-JOINTS - HARD MACHINING	6.000,00 €
1.2.31	1	WORKSHOP-ORIENTATED PROGRAMMING CUJ-JOINTS - HARD MACHINING	6.000,00 €
1.2.33	1	SERIES TRANSFORMER - MEXICO	3.750,00 €
1.2.34	1	AUXILIARY COOLING SYSTEM	3.500,00 €
1.2.36	1	4-COLOUR STATUS LIGHT (Q-LIGHT)	1.200,00 €
1.2.37	1	PNEUMATIC PRESSURE BOOSTER	1.000,00 €
1.2.38	1	AIR DRYER FOR COMPRESSED AIR SUPPLY	5.800,00 €
1.2.39	1	SPECIAL MACHINE COLOUR	3.450,00 €

1.3 COMPONENT RELATED EQUIPMENT

1.3.2	1	SET OF TURNING TOOL HOLDER (HARD) - BJ#22, BJ#23 & CUJ#27	3.450,00 €
1.3.3	1	SET OF TURNING TOOLS (HARD)	5.800,00 €
1.3.4	1	SET OF HARD MILLING TOOLHOLDERS HSK 63-C - BJ#22	5.550,00 €
1.3.5	1	SET OF HARD MILLING TOOLHOLDERS HSK 63-C - BJ#23	5.550,00 €
1.3.6	1	SET OF HARD MILLING TOOLHOLDERS HSK 63-C - CUJ#27	5.550,00 €
1.3.7	1	SET OF MILLING TOOLS FOR HARD MILLING	24.900,00 €
1.3.8	1	TEST BARS HSK63-C AND CAPTO C6	2.500,00 €
1.3.9	1	SETTING DIE FOR MILLING HEAD EXCHANGE	825,00 €
1.3.10	1	EQUIPMENT FOR BALL TRACK MILLING MACHINE	1.950,00 €
1.3.12	2	ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER SIN85 - with 1 medium duct	10.500,00 €
1.3.13	2	AIR SENSING	10.800,00 €
1.3.16	2	COLLET CHUCK (EMUGE) - ExaClamp	23.200,00 €
1.3.17	2	SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - BJ#22	10.000,00 €
1.3.18	2	SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - BJ#23	10.000,00 €
1.3.19	2	SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - CUJ#27	10.000,00 €

1.4 SERVICES

1.4.2	1	PROJECT ENGINEERING	37.100,00 €
1.4.3	1	TOOLING LAYOUT FOR OUTER RACE BJ#22 - HARD	8.700,00 €
1.4.4	1	TOOLING LAYOUT FOR OUTER RACE BJ#23 - HARD	8.700,00 €

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1.4.5	1	TOOLING LAYOUT FOR OUTER RACE CUJ#27 - HARD	8.700,00 €
1.4.6	1	MACHINE SETUP FOR OUTER RACE BJ#22 - HARD	12.500,00 €
1.4.7	1	MACHINE SETUP FOR OUTER RACE BJ#23 - HARD	12.500,00 €
1.4.8	1	MACHINE SETUP FOR OUTER RACE CUJ#27 - HARD	12.500,00 €
1.4.9	1	PRE-ACCEPTANCE OF THE MACHINE - HARD MACHINING	9.200,00 €
1.4.10	1	PRE-ACCEPTANCE OF THE MACHINE - HARD MACHINING	9.200,00 €
1.4.11	1	FUNCTION TEST OF THE MACHINE - HARD MACHINING	3.350,00 €
1.4.12	1	INSTALLATION AND DEMONSTRATION (OVERSEAS)	35.000,00 €
1.4.13	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR BJ#22 - HARD	5.500,00 €
1.4.14	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR BJ#23 - HARD	5.500,00 €
1.4.15	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR CUJ#27 - HARD	5.500,00 €
1.4.16	1	PRODUCTION SUPPORT AFTER START OF PRODUCTION	12.250,00 €
1.4.21	4	ADDITIONAL SET OF DOCUMENTATION	4.000,00 €
1.4.22	3	DOCUMENTATION ON USB	375,00 €
1.4.26	1	COMPLIANCE SPECIFICATION SHEET / DELIVERY CONDITIONS	11.300,00 €
1.4.28	1	OPERATING COURSE (Basic training)	6.125,00 €
1.4.29	1	MAINTENANCE (MECHANICAL AND ELECTRICAL) SERVICE COURSE (Basic training)	6.125,00 €
1.4.31	1	PACKING AND TRANSPORTATION OF TEST PIECES	4.800,00 €
1.4.32	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGES - KOR --> GER	10.200,00 €
1.4.33	1	PACKING OF INSPECTION GAUGES AFTER FINAL ACCEPTANCE	1.000,00 €
1.4.34	1	FOB HAMBURG	30.700,00 €
Total price:			1.308.225,00 €
Unique special discount:			-378.225,00 €

1.5 MACHINE SPECIFICATIONS

Final price: **930.000,00 €**

Price does not include options (*)

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1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 DUO KBU

1.1 BASIC MACHINE

- 1.1.1 1 MINERALIT (CAST GRANITE) MACHINE BED
consisting of two separate sections, to prevent interaction during the machining process. For transport purposes the two sections are mechanically coupled.
The machine bed is manufactured from high quality Mineralit (cast granite) which guarantees superior thermal stability and excellent dampening characteristics. Mineralit has excellent vibration dampening properties which are six to eight times better than cast iron. The large distance between the two high precision pre-stressed linear roller guideways of the X-axis guarantees high accuracy and high dynamics. The guideways are outside the working area and therefore require no protective covers.
- 1.1.2 1 ENERGY CONTAINER WITH CONTROL CABINET
Electrical cabinet, cooled with water/air heat exchanger, equipped to VDE 0113 and EN 13849 and integrated in the over-all machine frame, thus ensuring a totally transportable unit. 400 V / 50 Hz machine voltage.
Other voltage on request.
The central grease supply system for lubrication of the recirculating ballscrews and guideway systems and also the service unit for compressed air supply are integrated in the energy container.
- 1.1.3 1 CNC-CONTROL SIEMENS
According to control specification.
- 1.1.4 1 HYDRAULIC UNIT
The hydraulic unit is equipped with a self regulating pump.
Supply 15 l/min (at 70 bar)
Tank capacity 63 l
- 1.1.5 1 COOLING UNIT (15 KW, water - air)
A cooling unit controls the thermal stability of the machine. Spindle bearings, drive unit and motor spindle for milling head are maintained at ambient temperature (10°C - 40°C).
Cooling capacity: 15 kW
- 1.1.6 1 MACHINE ENCLOSURE
The machining area is effectively separated from the load/unload area by the two side walls of the Mineralit machine base and a sliding door to the pick-up station.
An additional bellows completes the working area when loading and unloading the main spindle. Vapors generated during processing can't get into the environment.
A large door at the front of the machine provides access to the machining area (one for each spindle). The doors are equipped with a safety glass window to DIN EN ISO 23125:2010 specification.
Two inspection lamps are installed in each machining area (one at the B-axis and one at the spindle headstock cover traveling in X)

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- 1.1.7 2 **MULTI-FUNCTIONAL CROSS-SLIDE WITH Y-AXIS**
For the complete machining with driven tools or with drilling, milling, reaming or grinding spindles.
Large stroke of the Y-axis.
This unit is equipped with three axes (X, Y and Z) for each spindle. Both spindles can be programmed independently via the CNC control.
High rapid feed enable short idle times by component load/unload and gauging (optionally).
The axes drives are quick reaction frequency controlled service free servomotors which drive the slides over high precision ballscrew spindles. The X-, Y- and Z-axis are equipped with completely enclosed linear measuring systems.
On activation of the emergency-stop or in case of power failure, a brake will stop the vertical axis.
- 1.1.8 2 **MOTOR SPINDLE**
Spindle headstock unit with motor spindle
The motor spindle is a high dynamic frequency controlled service free AC synchronous spindle motor.
High degree of spindle rigidity due to triple bearing support over a short distance.
Precision shoulder bearings arranged in tandem (double-O). Additional bearing at spindle end for length compensation. All bearings are lifetime lubricated and thus maintenance free.
Sealing air to prevent dirt from penetrating from the outside.
The guideway system in the Z axis is equipped with hydrostatic bearings for compensation the cutting forces which guarantee excellent guideway quality by minimum friction.
Thermo-symmetrical design and integrated coolant system guarantee constantly maintained accuracy.
Positioning accuracy of spindle: +/- 0,01 angular degrees (C-axis)
Electro-hydraulic spindle locking optionally.
- 1.1.9 2 **MILLING HEAD FOR MILLING OF BALL GROOVES**
with 1 milling spindle
consisting of:
one axis, NC controlled swivelling unit (B-axis)
Swivelling housing guided by roller bearings in the side walls of the machine base.
Swivelling angle -30° to +90° (120°)
Swivelling drive by torque drive. Direct measuring system.
Mounted on the right side
- 1 MOTOR SPINDLE HPC190 (manually tool clamping)
- | | | |
|-----------------------------|-----|-----------|
| Max. power 40 % ED | kW | 15 |
| Torque at 40 % ED | Nm | 67 |
| Max. spindle speed | rpm | 8.500 |
| Full power at spindle speed | rpm | 3.000 |
| Spindle taper DIN 69893 | HSK | C63 / F80 |
- Fluid-cooled spindle housing. Hybrid bearing.
Lifetime lubricated. Temperature controlled.
Rotating components are precision-balanced in two planes.
Coolant supply to tool holder pocket with minimal quantity lubrication max. 10 bar.
Pneumatic spindle nose cleaning device. Air seal between spindle and housing.
(warranty period 18 months)

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1.1.10	1	EMAG OPERATING AND SETTING MASKS The EMAG masks offer the setter/ operator optimal support in setting and operating the machine. Multiple base- and monitoring functions including data input are, depending on the machine type, available.	
1.1.11	1	For simple and fast programming of the machine on an appropriate ball track, the machine can be equipped with workshop-oriented programming modules. These modules are not included in the basic machine price and are described separately in the chapter „options to the CNC control“. On the other hand, the part programs can also be completed by the operator in normal ISO code.	
1.1.12	1	STANDARD COLOUR Machine - light grey to RAL 7035, front door - grey to RAL 7046, automation, swarf conveyor - grey to RAL 7046.	
1.1.13	1	NOISE LEVEL A maximum average corrected sound level of 78 dB(A) to DIN 45635 is obtained when turning with a continuous cut on a C45 steel disc in both longitudinal and transverse direction. If the workpiece has for example an interrupted cut due to its geometry or its clamping condition there may occur a higher sound level. For an extra cost sound deadening material can be fitted to reduce this higher sound level.	
1.1.14	1	PRICE OF MACHINE VSC 315 DUO KBU (SIEMENS) - HARD MACHINING	654.700,00 €

1.2 ADDITIONAL EQUIPMENT

1.2.1 TOOLING SYSTEM

1.2.2	2	TOOL BLOCK FOR 1 TURNING TOOL CAPTO C6 manually clamped consisting of: toolholder with clamping insert C6 With internal air blast feeding to the tools. The coolant system guarantees constantly maintained accuracy.	30.800,00 €
1.2.3	2	PROTECTIVE PLATE FOR THE EMPTY TURNING TOOL POSITION Cooling is prepared for retrofitting tool holders	250,00 €

Price does not include options (*)

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**1.2.4 COMPONENT STORAGE AND TRANSPORT SYSTEM**

- 1.2.5 1 WORKPIECE LOADING by motorised, chain-type conveyor 22.200,00 €
Workpiece diameter: 35 - 120 mm
Length: approx. 2300 mm
Conveyors mounted to left of the machine.
Quick-change facility to adjust loading device to different workpiece diameters.
1x pneumatic stopper / separator ahead curve, adjustable.
Light beams.
1x workpiece stop.
Electric / pneumatic installation on plug distributor.
- 1.2.6 1 WORKPIECE STOP OUTER RACE / DISC TYPE
Stop for Outer Race and disc type OR
Workpiece height > 35 mm
- Please note: Included in Pos. 1.2.5.**
- 1.2.7 1 LIGHT BEAMS FOR LOADING CONVEYOR 950,00 €
1x storage position free / 1x loading conveyor full
For external loading
- 1.2.8 1 LINEAR GANTRY LP3, WITH ROTATION MODULE 23.300,00 €
2-axis linear gantry
Drive by NC servomotor. Direct connection to the machine.
Energy guiding on all axes via cable track.
Rotation unit driven by a servo motor with rotary inlet.
Workpiece transport with a pneumatic 3-finger centric gripper
incl. 1 set of workpiece-specific gripper jaws.
without position detection and sensor gripper monitoring.
Prepared for fitting of an automatic alignment device for the correct angular positioning of
workpieces in the ball tracks (hard machining).
- Swivel angle $\pm 360^\circ$
X-axis stroke 630 mm
Z-axis stroke 500 mm
Workpiece weight max. 3,5kg
- 1.2.9 1 GRIPPER JAWS OUTER RACE CLAMPING DIA. 20-36
1 set of gripper jaws for 3-finger centric part gripper
- Included in pos. "LINEAR GANTRY LP3, WITH ROTATION MODULE"**
- 1.2.10 1 POSITION DETECTION 2.550,00 €
Position detection of the workpiece via Keyence laser sensor, mounted on the gantry
frame LP3
- 1.2.11 1 NIO - CHUTE, LENGTH 1550mm, R-L / L-R 3.300,00 €
Unloading, flow direction R-L / L-R
Design as a chute with structural sheet metal
For non-oriented or wrong workpieces, consisting of:
Direction to the side $\rightarrow$ right / left
1x light beam
Length: approx. 1550 mm
Accumulation range: approx. 1350 mm

Price does not include options (*)

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1.2.12	1	SHUTTLE ZR120DUO L-R/ L-L FOR WORKPIECE LOADING Type: ZR120DUO-A-1600-1305-800-LE-B-1 For the workpiece transport through the machine, with a basic pallet for blank parts. Consists of a spring-loaded pick-up station, including detection and emergency stop. Slip / brake plate coated with spring steel, 1 through for dry processing. Mounting kit on the machine base body, 1x light barrier. Driven via servomotor. The pick-up spindle grasps the blank parts of the shuttle. The workpieces are positioned NC-controlled for the pick-up spindle. Different workpiece diameters and heights are defined in the NC part program.	40.700,00 €
1.2.13	2	WORKPIECE CARRIER RIGID OUTER RACE BJ#22 Hard machining - raw part for the location of outer races. Designed for one workpiece.	3.700,00 €
1.2.14	2	WORKPIECE CARRIER RIGID OUTER RACE BJ#23 Hard machining - raw part for the location of outer races. Designed for one workpiece.	3.700,00 €
1.2.15	2	WORKPIECE CARRIER RIGID OUTER RACE CUJ#27 Hard machining - raw part for the location of outer races. Designed for one workpiece.	3.700,00 €
1.2.16	2	AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK For the cleaning of the pick-up unit and the clamping chuck by air blast. The air is processed by a service unit and directed at the component through nozzles.	2.400,00 €
1.2.17	1	WORKPIECE UNLOADING by motorised, chain-type conveyor Workpiece diameter: 35 - 120 mm Length: approx. 5000 mm Conveyors mounted to right of machine. Quick-change facility to adjust loading device to different workpiece diameters. 1x pneumatic stopper / separator ahead curve, adjustable. Light beams 1x workpiece stop. Electric / pneumatic installation on plug distributor.	28.500,00 €
1.2.18	1	LIGHT BEAM FOR UNLOADING CONVEYOR 1 x workpiece at the unloading position For external unloading	900,00 €

Price does not include options (*)

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Managing Directors:
Dr. Guido Hegener, Dr. Mathias Klein
Commercial Register Ulm: HRB 533796
Tax No.: 63002/12866 FA Göppingen
VAT No.: DE 813 898 840

Bank accounts:
UniCredit Bank AG / Hypovereinsbank Stuttgart 'IBAN DE02 6002 0290 0322 6388 17' BIC HYVEDEMM473
Commerzbank AG Tübingen 'IBAN DE09 6414 0036 0890 9103 00' BIC COBADEFF641
Kreissparkasse Göppingen 'IBAN DE96 6105 0000 0015 5885 93' BIC GOPSDE6GXXX
Bayerische Landesbank München 'IBAN DE59 7005 0000 0004 2569 67' BIC BYLADEMMXXX



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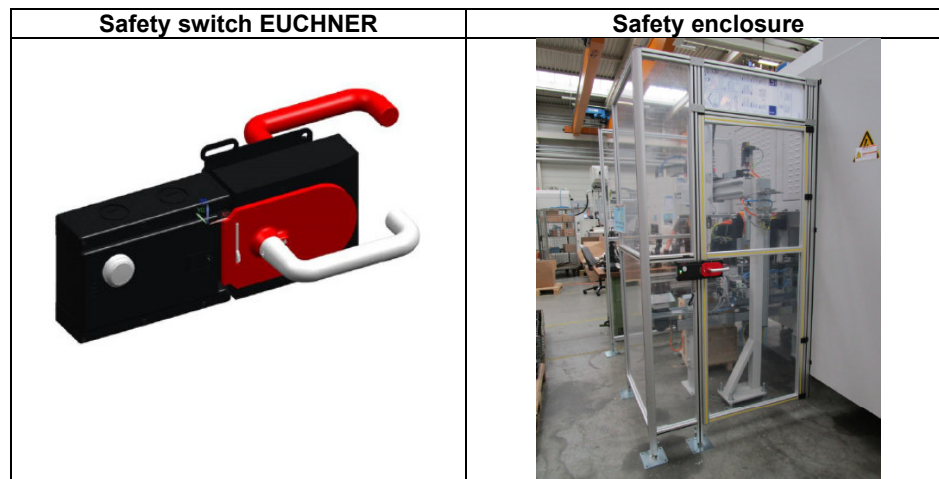
Austraße 24
73084 Salach - Germany
Phone: +49 7162 17-0
Fax: +49 7162 17-4270
E-Mail: info@emag.com
Internet: www.emag.com

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1.2.19 1 SAFETY ENCLOSURE 10.900,00 €

Design consisting of:

- Aluminium profile (size 40)
 - Height of the safety fence: = 2400 [mm]
 - Space between ground and safety fence: 180 [mm]
 - Corner post quarter-round, all others square
 - Inserts Makrolon-discs, no woven wire mesh
 - Safety fence painted as follows:
 - Groove covers in door: yellow (RAL 1018)
 - Groove covers other fields: black
 - Makrolon discs: transparent 5mm thick
 - Visible profile ends covered with black plastic caps.
 - 3x Safety switch MGB Set EMAG, type MGB-L1HE-AR-R/L-EMAG, Maker: EUCHNER
- Enclosure covering moving automation components. Four access doors mechanically and electrically lockable.



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EMAG Maschinenfabrik GmbH

Austraße 24
73084 Salach - Germany
Phone: +49 7162 17-0
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EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

1.2.20 COOLANT AND SWarf DISPOSAL SYSTEM

- | | | | |
|--------|---|---|-------------|
| 1.2.21 | 1 | 2 SWarf CONVEYORS, CHAIN TYPE
Consisting of two single swarf conveyors:
The first swarf conveyor is mounted laterally on the unloading side of the machine.
Swarf exit height 350 mm
The second swarf conveyor carries the swarfs to the rear side of the machine.
Swarf exit height 1250 mm. | 27.400,00 € |
| 1.2.22 | 2 | CENTRAL EXTRACTOR UNIT (PREPARATION)
Connector for central extractor unit, mounted at the machining area.
Diameter 150 mm. Required extractor capacity approx. 1000 m ³ /h

Supply of a suitable extractor unit, for setup and pre acceptance of the machine at manufacturer's plant, by the customer or of EMAG at extra costs. | 1.650,00 € |
| 1.2.23 | 1 | CHIP BOX (LARGE VERSION)
with caster | 2.750,00 € |
| 1.2.24 | 1 | AIR GUN
with hose in blue color. | 1.050,00 € |



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1.2.25	OPTIONAL EQUIPMENT FOR CNC CONTROL		
1.2.26	1	NC-GUIDED EMERGENCY RETRACTION AT POWER FAILURE (ESR) Interpolated retraction of axis in case of power failure, voltage fluctuation and drive failure. Electric components (condenser modul) are used which maintain the voltage for the duration of the retreat and use the energy for the nc-guided retract of the axes. (strongly recommended for unstable power supply, expensive tools or workpieces)	10.500,00 €
1.2.27	1	NETWORK DRIVE CONNECTION (Control system Siemens 840Dsl) Linking the machine control to your network after startup in your factory, so that network drives are available in the user interface. NC-programs or any data can be stored or loaded from there. Customer-specific requirements / restrictions for network integration: - Operation with DHCP server yes / no; if no, then IP address, sub-net mask and default gateway - DNS server and enable code of network source (also IP-address of server, if DNS not known) - User name and password for the automatic preparation of the connection - Network cable into the electrical cabinet When providing server and directory names it has to be taken into account that the 840Dsl features an English language Windows operating system that cannot handle the German umlauts ä, ö and ü. When installing of security update, the company Siemens restrictions are to consider. If in the domain automatically software is installed on all computers, so the functionality is no longer guaranteed after restart!	2.000,00 €
1.2.28	2	INTERFACE TO EXTERNAL HANDLING SYSTEM - HARDWARE Safety signal interface via HAN24E plug on electrical cabinet with 8 potential-free inputs and 8 potential-free outputs for the change of safety signals "emergency-stop" and "access door open" Handling-signal interface via HAN16E plug on electrical cabinet Deviations of the standard interface on request at additional costs. (1.000,00 € / signal)	7.400,00 €
1.2.29	1	CONNECTING KIT FOR MINI HAND-HELD CONTROL UNIT – SIEMENS	825,00 €
1.2.30	1	WORKSHOP-ORIENTATED PROGRAMMING AC-JOINTS - HARD MACHINING For outer races (straight grooves). For quick and easy programming parameters for componentrelated dimensions are adapted. Same process for tool-offset. Every tool is allocated an empirically determined preset quantity of workpieces to be machined. On expiry of the tool life the machine is stopped. Control monitor display: tool-no., preset and remaining tool life quantity of workpieces per tool.	6.000,00 €

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EMAG Maschinenfabrik GmbH

Austraße 24
73084 Salach - Germany
Phone: +49 7162 17-0
Fax: +49 7162 17-4270
E-Mail: info@emag.com
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1.2.31	1	WORKSHOP-ORIENTATED PROGRAMMING CUJ-JOINTS - HARD MACHINING For outer races (straight grooves). For quick and easy programming parameters for componentrelated dimensions are adapted. Same process for tool-offset. Every tool is allocated an empirically determined preset quantity of workpieces to be machined. On expiry of the tool life the machine is stopped. Control monitor display: tool-no., preset and remaining tool life quantity of workpieces per tool.	6.000,00 €
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EMAG Maschinenfabrik GmbH

Austraße 24
73084 Salach - Germany
Phone: +49 7162 17-0
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EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

1.2.32 COUNTRY SPECIFIC EQUIPMENT

1.2.33	1	SERIES TRANSFORMER (63A) (low version) for 3-phase power supply, with neutral conductor and ground conductor. Power supply (prim): 380, 420, 440, 460, 480 V - 50 / 60 Hz Other supply on request. Transformer in separate housing. Located on the machine. Customer pre-fuse required. Connection and link wiring provided by customer.	3.750,00 €
1.2.34	1	AUXILIARY COOLING SYSTEM consisting of auxiliary coolant supply unit for switchgear cabinet.	3.500,00 €

Price does not include options (*)

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**1.2.35 OTHER AUXILIARY EQUIPMENT**

1.2.36	1	4-COLOUR STATUS LIGHT (Q-LIGHT) mounted on machine. Colour code: red: machine stopped yellow: warning -tool to be changed, -lubricant running low, etc. green: machine in automatic cycle. white: M/C ready (waiting for workpiece supply)	1.200,00 €
1.2.37	1	PNEUMATIC PRESSURE BOOSTER to increase air pressure from 4 bar to 6 bar. (only for part grippers) Maker: Festo Type: DPA-63-10	1.000,00 €
1.2.38	1	AIR DRYER FOR COMPRESSED AIR SUPPLY Absorption air dryer for dry and clean air. Manufacturer: Festo Type: PDAD-51-G1/2 incl. combined maintenance unit	5.800,00 €
1.2.39	1	SPECIAL MACHINE COLOUR Complete machine according to customers colour specification.	3.450,00 €

1.3 COMPONENT RELATED EQUIPMENT**1.3.1 TOOL EQUIPMENT**

1.3.2	1	SET OF TURNING TOOL HOLDER (HARD) - BJ#22, BJ#23 & CUJ#27 consisting of: - 2 turning tools Capto C6 - on the MC - 2 turning tools Capto C6 - as spare	3.450,00 €
1.3.3	1	SET OF ROUGH TURNING TOOLS (HARD) consisting of: - 40 cbn cutting inserts	5.800,00 €
1.3.4	1	SET OF MILLING TOOLHOLDERS HSK 63-C - BJ#22 Toolholders for interchangeable milling heads. - 2 pcs milling holder for BJ#22 (2x on the MC) - 2 pcs milling holder for BJ#22 (2x as spare)	5.550,00 €
1.3.5	1	SET OF MILLING TOOLHOLDERS HSK 63-C - BJ#23 Toolholders for interchangeable milling heads. - 2 pcs milling holder for BJ#23 (2x on the MC) - 2 pcs milling holder for BJ#23 (2x as spare)	5.550,00 €
1.3.6	1	SET OF HARD MILLING TOOLHOLDERS HSK 63-C - CUJ#27 Toolholders for interchangeable milling heads. - 2 pcs milling holder for CUJ#27 (2x on the MC) - 2 pcs milling holder for BCUI#27 (2x as spare)	5.550,00 €

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1.3.7	1	SET OF MILLING TOOLS FOR HARD MILLING Head system with pcbn cutting inserts - 30 cutting heads Ø 16.0[mm] for BJ#22 - 30 cutting heads Ø 16.8[mm] for BJ#23 - 30 cutting heads Ø 20.3[mm] for CUJ#27	24.900,00 €
1.3.8	1	TEST BARS HSK63-C AND CAPTO C6 for checking the machine geometry Consisting of: 1x aluminum case 1x foam insert 1x test bar HSK63-C 1x test bar Capto C6	2.500,00 €
1.3.9	1	SETTING DIE FOR MILLING HEAD EXCHANGE consisting of : 1x fixture for milling head change for mounting on a bench.	825,00 €
1.3.10	1	EQUIPMENT FOR BALL TRACK MILLING MACHINE consisting of: - 1 cone wiper - 1 torque wrench with insert - 2 plugs HSK63 - 1 ball plug gauge HSK63	1.950,00 €

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**1.3.11 CHUCKING SYSTEM**

- 1.3.12 2 ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER SIN85 10.500,00 €
- with 1 medium duct
Max. drawbar force 3.325 daN at 70 bar.
The clamping pressure is manually infinitely adjustable. .
The clamping cylinder stroke and the clamping pressure are controlled by a linear measuring system and by a pressure switch.
- 1.3.13 2 AIR SENSING 10.800,00 €
The air sensing is exclusively used to identify the workpiece contact on the clamping device.
This is done by measuring the dynamic pressure that occurs between the workpiece and the clamping device.
The air is supplied by one or more nozzles (depending on clamping fixture type).
The pressure is measured with a pressure switch.
The air supply is controlled by a directional valve.
The air flow is sent through the main spindle.

NOTE:

Requires CHUCKING CYLINDER WITH MEDIUM DUCT

Equipping the clamping fixture with a pneumatic system monitor is not included in the price of this item.

1.3.14 COLLET CHUCK - DIRECT CONNECTION**COLLET CHUCK EMUGE ExaClamp**

- 1.3.15 1 COLLET CHUCK EMUGE ExaClamp
Please note that the following EMUGE collet chuck is a new standard chuck.
The clamping heads / collets from the older version are compatibel with the new standard chuck.
The part stops / datum plates and dirt covers are not compatibel.
- 1.3.16 2 COLLET CHUCK (EMUGE) - ExaClamp 23.200,00 €
For clamping of Outer Races and Outer Race Discs.
Installation direct on the main spindle (no spindle adapter required).
With basic part stop.
Clamping range: 65 mm - 120 mm
Prepared for air sensing.
Without workpiece-specifically interchangeable parts.
- 1.3.17 2 SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - BJ#22 10.000,00 €
Laid out for one Outer Race.
Consisting of clamping head, part stop, dirt cover and in total one control ring.
- 1.3.18 2 SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - BJ#23 10.000,00 €
Laid out for one Outer Race.
Consisting of clamping head, part stop, dirt cover and in total one control ring.
- 1.3.19 2 SET OF INTERCHANGEABLE PARTS (COLLET CHUCK) - CUJ#27 10.000,00 €
Laid out for one Outer Race.
Consisting of clamping head, part stop, dirt cover and in total one control ring.

1.4 SERVICES

Price does not include options (*)

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1.4.1	1	Please note that no additional services are contained in the previously specified prices. We would be pleased to carry out the following services and would quote according to your wishes. - Preparation of tooling layouts and NC programs - Set-up machine for component trials on your specific workpiece in our works - Machine approval in our works by your engineers. - Final acceptance at customer's plant with a specific workpiece	
1.4.2	1	PROJECT ENGINEERING Project engineering and coordination with suppliers and customers. Engineering investigations (machine set up, with central automation, central exhaust, chip disposal) Customization of interfaces with the suppliers.	37.100,00 €
1.4.3	1	TOOLING LAYOUT FOR OUTER RACE BJ#22 - HARD Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.	8.700,00 €
1.4.4	1	TOOLING LAYOUT FOR OUTER RACE BJ#23 - HARD Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.	8.700,00 €
1.4.5	1	TOOLING LAYOUT FOR OUTER RACE CUJ#27 - HARD Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.	8.700,00 €
1.4.6	1	MACHINE SETUP FOR OUTER RACE BJ#22 - HARD Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	12.500,00 €
1.4.7	1	MACHINE SETUP FOR OUTER RACE BJ#23 - HARD Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	12.500,00 €
1.4.8	1	MACHINE SETUP FOR OUTER RACE CUJ#27 - HARD Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	12.500,00 €
1.4.9	1	PRE-ACCEPTANCE OF THE MACHINE - HARD MACHINING Specific workpiece: Outer Race BJ#22 with measuring equipment supplied by customer. One workpiece in one clamping operation. Max. 30 components. Submission of proof to Cm of machine's capability of max. 3 measuring characteristics. For Cm of machine's capability measurements of workpiece 1, 5, 10, 15, 20, 25, 30.	9.200,00 €
1.4.10	1	PRE-ACCEPTANCE OF THE MACHINE - HARD MACHINING Specific workpiece: Outer Race CUJ#27 with measuring equipment supplied by customer. One workpiece in one clamping operation. Max. 30 components. Submission of proof to Cm of machine's capability of max. 3 measuring characteristics. For Cm of machine's capability measurements of workpiece 1, 5, 10, 15, 20, 25, 30.	9.200,00 €

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1.4.11	1	FUNCTION TEST OF THE MACHINE - HARD MACHINING Specific workpiece: Outer Race BJ#23 for one workpiece including test run, without SPC analysis, 10 parts machined in 100% drawing tolerance. With measuring equipment supplied by customer.	3.350,00 €
1.4.12	1	INSTALLATION AND DEMONSTRATION (OVERSEAS) in your works The following services are included: - Machine installation and demonstration - Machine approval and hand-over - Talk-through of the machine documentation and service instruction - Operator instruction - Travel costs Total of 15 man days Not included are: - Machine set-up - Machine operation for production - Additional service Additional service engineer costs for time in excess of that quoted will be charged extra.	35.000,00 €
1.4.13	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR BJ#22 - HARD Specific workpiece: Outer Race BJ#22 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces Evaluation and guaranteed criteria are the same as for pre-acceptance test.	5.500,00 €
1.4.14	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR BJ#23 - HARD Specific workpiece: Outer Race BJ#23 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces Evaluation and guaranteed criteria are the same as for pre-acceptance test.	5.500,00 €
1.4.15	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT OR CUJ#27 - HARD Specific workpiece: Outer Race CUJ#27 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces Evaluation and guaranteed criteria are the same as for pre-acceptance test.	5.500,00 €
1.4.16	1	PRODUCTION SUPPORT AFTER START OF PRODUCTION 1 week (6 days - monday - saturday) production support at customers plant. Performed by EMAG Mexico.	12.250,00 €
1.4.17	1	INSTALLATION POINT The installation point has to be prepared in accordance with the documentation supplied. We take it for granted that all conditions are met before our installation crew arrives. Where waiting times are incurred due to the conditions not having been met, these will be charged extra. During installation and commissioning our engineers have to be provided, on request, with the support of skilled customer personnel and similar assistance. This also applies to the operation of cranes, handling equipment and other aids.	

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EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

- 1.4.18 1 **MACHINE LIFTING EQUIPMENT**
EMAG will provide a lifting equipment to lift the machine by crane, free of charge.
However, should the customer not return the equipment to EMAG within 4 weeks, freight
and charges prepaid, a separate invoice of EUR 1.500,00 will be issued.

Price does not include options (*)

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EMAG Maschinenfabrik GmbH

Austraße 24
73084 Salach - Germany
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EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

1.4.19 DOCUMENTATION

1.4.20 1 DOCUMENTATION (Siemens)

OPERATING INSTRUCTIONS

1 complete set of operating instructions on paper

EU countries in the official language of the country; other countries on national regulations

Manuals for the machine:

- Introduction
- Safety instructions
- Product description
- Transport and installation
- Start-up
- Production
- Preventive maintenance

Manuals from the manufacturer of the control unit:

Siemens SIN 840D sl

- User manual
- Diagnostics manual
- Programming - Basics
- Programming - Cycles
- Programming - Operations planning and scheduling

Machine diagrams:

- Machine layout
- Transport diagram
- Machine data sheet
- Drive data sheet
- Connection data sheet for fluidics
- Connection data sheet for electrics

List of spare parts and wearing parts

Operating instructions from third parties

Documentation for clamping devices and tools

TECHNICAL DATA

1 set of technical data on paper, German or English

- Electrical circuit diagram with list of el. devices
- Fluid system diagram with list of el. devices
- Drawings of subassemblies with relevance for spare parts
- List of components with relevance for spare parts and wearing parts in relation to the drawings
- Technical data, test reports, declarations of incorporation from third parties

Other languages available on request for an additional charge; third party documentation excepted

1.4.21	4	ADDITIONAL SET OF DOCUMENTATION 1 additional set of the documentation described above. Language: 3x Spanish, 1x Korean	4.000,00 €
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Price does not include options (*)

Offer : 196564 / 5

Managing Directors:
Dr. Guido Hegener, Dr. Mathias Klein
Commercial Register Ulm: HRB 533796
Tax No.: 63002/12866 FA Göppingen
VAT No.: DE 813 898 840

HYUNDAI WIA Corporation

Bank accounts:
UniCredit Bank AG / Hypovereinsbank Stuttgart 'IBAN DE02 6002 0290 0322 6388 17' BIC HYVEDEMM473
Commerzbank AG Tübingen 'IBAN DE09 6414 0036 0890 9103 00' BIC COBADEFF641
Kreissparkasse Göppingen 'IBAN DE96 6105 0000 0015 5885 93' BIC GOPSDE6GXXX
Bayerische Landesbank München 'IBAN DE59 7005 0000 0004 2569 67' BIC BYLADEMMXXX

19/08/2022

22/32



EMAG Maschinenfabrik GmbH

Austraße 24
73084 Salach - Germany
Phone: +49 7162 17-0
Fax: +49 7162 17-4270
E-Mail: info@emag.com
Internet: www.emag.com

EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

1.4.22	3	DOCUMENTATION ON USB 1 complete set of documentation on USB Language: 2x English, 1x Spanish	375,00 €
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Price does not include options (*)

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1.4.23 CE-MARKING

- 1.4.24 1 WITHOUT CE MARKING / WITHOUT DECLARATION OF CONFORMITY (non-EU country, complete machine)
The machinery is always designed and constructed in accordance with the applicable harmonized standards and safety instructions. For operation of the machine, the regional regulations have to be observed additionally.

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1.4.25

CUSTOMER SPECIFICATION

1.4.26

1

COMPLIANCE SPECIFICATION SHEET / DELIVERY CONDITIONS

Design of mechanical-, electrical- and fluid equipment according to "M/C Specification, Date: 1st July 2022 - WGT-M4-0022-001 - 1st version"

Due to our design we are unable to comply with the following items of your specification sheet, meaning we must apply for an exceptional permission.

11.300,00 €

Item	Customer requirement	EMAG commitment/Exceptional approval
Basic Condition	Surrounding temperature: 0°C - 40°C	Surrounding temperature: +10°C to +40°C
Noise level	less than 75 dB (A)	up to 78dB (A)
General specification of Hyundai WIA	Keep the "General specification of Hyundai WIA"	This document is not part of the inquiry and therefore it is not considered.
Main Circuit	Solenoid valve: 110V AC 60Hz, Lamp Type, Communication Type	24V DC with LED
CNC & PLC	- Voltage 240V - Memory USB Card - Memory EPROM	400V AC / 24V DC - Memory USB-Stick - No longer available from Siemens.
CNC & PLC	Network interface - LAN	USB
Wiring	Remain the 30% marginal space and do not wire it hose together	max. 10% possible
Items for the capability (pre- and final acceptance)	Ball Track Milling: Groove Dia, Center, Ball Diameter	Ball diameter is no item for evaluation (Cm)
Detail 14. Others (Lub) &	2) Hyd. Pump - The type of hyd. Pump should be variable discharge 4) Inlet The mesh could be #100-120 mesh 2) Maker should install the air dryer to remove the water. Specification of air dryer: SMC-IDF15E1 14) The chuck maker should be EMUGE (for common use for existing line) The pallet for raw part should be pin type. Neither ball nor chuck type is allowed. 33) LMS should be established and connect to MES. "Work name, Status (machining, waiting, Stop), counter, feedrate, C/Time etc. 36) When SPC time comes, m/c alarm after discharge of the workpiece, and then starts automatically after SPC is done.	2) Pump engine is frequency controlled. 4) EMAG use 10µm 2) EMAG use FESTO membrane air dryer MS6-LDM1-1/2-P20. Please either grant us an exemption or provide us with more details about your requested air dryer SMC-IDF15E1 14) EMUGE has a new standard collet ExaClamp. Only the clamping collets are compatible. The datum's/ part stops and dirt covers are not compatible. The pallet for raw part (soft machining) are spring actuated ball clamps. Same design as delivered with MC SN-108818. The tolerance of the preforged ball tracks is too large for rigid workpiece carriers  33) Same functionality and scope as in the delivered machines P-30112294/SN108817 and P-30112295/SN108818. No individual tracking of workpieces considered. 36) The operator must manually confirm after removing and checking the SPC workpieces.
Safety Fence	According to WIA Specification.	Please refer to pos. 1.2.19 in this offer. For some items an exception approval is required.
Electric	OP Panel: Height = FL + 1300mm	Height of the OP Panel: Keys approx. 1200mm to 1440mm Screen approx. 1450mm to 1750mm

Price does not include options (*)

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EMAG Maschinenfabrik GmbH • P O Box 1152/ 1162 • 73080 Salach • Germany

1.4.27 TRAINING

- | | | | |
|--------|---|---|------------|
| 1.4.28 | 1 | OPERATING COURSE (Basic training)
For two participants in your works, performed by EMAG Mexico.
Matched to the machine design (modular customized)
Training course duration: 3 days / shift (shift monday - saturday).
Language: Spanish
Documentation: 4x Spanish | 6.125,00 € |
| 1.4.29 | 1 | MAINTENANCE (MECHANICAL AND ELECTRICAL) SERVICE COURSE (Basic training)
For two participants in your works, performed by EMAG Mexico.
Matched to the machine design (modular customized)
Training course duration: 3 days / shift (shift monday - friday).
Language: Spanish
Documentation: 4x Spanish | 6.125,00 € |

Price does not include options (*)

Offer : 196564 / 5

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1.4.30 TRANSPORT COSTS

1.4.31	1	PACKING AND TRANSPORTATION OF TEST PIECES Transport (ocean freight) from customers plant in Mexico or Korea to EMAG Maschinenfabrik in Salach. - 300 test pieces of acceptance part - 150 test pieces for the 1st function test part - 150 test pieces for the 2nd function test part. Max. total weight = 950 [kg].	4.800,00 €
1.4.32	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGES - KOR --> GER Transport from customers plant in Korea to EMAG Salach Germany. (Air freight, max. 750kg)	10.200,00 €
1.4.33	1	PACKING OF INSPECTION GAUGES AFTER FINAL ACCEPTANCE Packing of the inspection gauges after final acceptance by EMAG employees. Responsible for seaworthy packing is the customer Hyundai WIA. It must be ensured that EMAG can reuse the seaworthy packaging.	1.000,00 €
1.4.34	1	FOB HAMBURG from Salach, Incoterms 2020	30.700,00 €
Total price:			1.308.225,00 €
1.4.35	1	UNIQUE SPECIAL DISCOUNT	-378.225,00 €

1.5 MACHINE SPECIFICATIONS

Price does not include options (*)

Offer : 196564 / 5

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1.5.1 1 TECHNICAL MACHINE DATA VSC 315 DUO KBU Subject to change without notice

WORKING AREA

Chuck diameter	mm	250
Swing diameter	mm	260
X-stroke	mm	900
Y-stroke	mm	315
Z-stroke	mm	300

WORKPIECE DIMENSIONS

Workpiece length, max.	mm	260*
Workpiece O/D, max.	mm	120
Ball section dia., max.	mm	30

(*Longer workpieces on request)

MAIN SPINDLE

Spindle flange acc. to DIN 55026	size	8
Spindle front bearing diameter	mm	140
max. spindle speed	rpm	3000

MAIN DRIVE

AC-Synchronous motor 40/100 % ED	kW	28,5/27,5
Max. power	kW	28,5
Full power at spindle speed	rpm	700
Torque at 40/100 % ED	Nm	590/375
Max. torque at	Nm	590

FEED DRIVE

Rapid traverse feed, X+Y	m/min	45
Rapid traverse feed, Z	m/min	30
Feed force, X/Y/Z	kN	11
Ball screw diameter X/Y/Z	mm	50/40/40

MILLING SPINDLE

AC-motor	kW	15
Max. spindle speed	rpm	8500

ELECTRICAL EQUIPMENT according to VDE 0113

Supply voltage	V	400
Control circuit voltage - DC	V	24
- AC	V	230
Frequency	Hz	50
Total connected load	kVA	120
Input fuse	A	250

DIMENSIONS AND WEIGHT

Length	mm	4400
Width	mm	3600
Height	mm	3500
Weight	kg approx.	20000

1.5.2 1 SIEMENS OPERATOR PANEL 15" MULTITOUCH MONITOR
consisting of OP015 Black + IPC 427E + Control panel (MCP483C)

Price does not include options (*)

Offer : 196564 / 5

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- 1.5.3 1 **CNC CONTROL SIEMENS**
SINUMERIK 840 Dsl with integrated PLC S7-300
Operator software Sinumerik Operate with HMI-Pro (Windows 10)
Screen language and on-screen messages in German / English / in official EU language,
more languages available as an optional
CNC program and data memory 10 Mbyte
USB-interface (front) for memory-stick (USB 2.0)
Machine diagnostics via internet with VPN technology to verify EMAG's standard format
Software backup is to external media (USB stick integrated into IPC) to verify EMAG's
standard format
dataGrabber (Comara) – Capable of recording process data for diagnostics, optional
process optimization
No active evaluation possible on the control, storage of the data takes place in a ring
buffer

EMAG ADD-ON FUNCTIONS:

Machine-specific documentation of functions and parameters is available directly on the
control unit. (PDF display)
Automatic return to the home position upon machine error. Error recovery is a simple one
button process.
First run high speed traverse is at a reduced speed from home to the first cut. Reduced
speed allows you to avoid a collision and save damaged equipment and reprogramming
costs.
Internal measuring probe with collision monitoring on the traversing movements (Option)
Automatic workpiece gaging when switching to a replacement tool or tool insert (Option)
Automatic filling and emptying of the entire machine.
"Optional Stop" (M01) allows you to make a programmed stop or interrupt production after
finishing a part at a desired point.
Single cycle: Allows one part to be transferred and machined for setup and tryout
purposes.

STRUCTURED WORKPIECE PROGRAMMING:

Division of individual programs into subprograms, e.g. the program for the machining
contains only the technology program and not the loading part.
The loading program has individual parameters. Division into machine and work piece
specific parameters.
Programmable chuck with length measuring system in the pull rod.
Different clamping situations (Unclamped, Clamped and Over stroke) can be changed by
parameter assignment. Displayed by diagnostics menu.
Recording and displaying of the most important production data: work piece counter, shift
counter, cycle time, work piece status, operating hours.
Production of definable batch sizes. The machine stops upon reaching the selected no.
Optimized programming for the turret keeps chips from collecting.
Rinsing of clamping devices after a programmable number of parts
Number and positions of the workpiece carriers can be defined for closed automation.
Lifecycle screen that displays tool wear and tool life.

Final price:

930.000,00 €

Price does not include options (*)

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Commercial Conditions

CONFIDENTIALITY

Any commercial and operational written or oral information or data relating to this quotation / order, such as drawings, specifications, recommended machining data and clamping methods, etc., shall not be divulged to any third party (referring, in particular, but not exclusively, to competitors, for the purpose of bid comparisons) and are to be used solely within the meaning of this quotation / order.

The recipient of this quotation / order expressly accepts these obligations.

MACHINING

The determined cycle times result from values, which are based on data and experiences of acknowledged cutting material manufacturers. Changing rough part dimensions, tolerances, material as well as the material hardness are only possible during building and testing of the machine. Changes will only be accepted in written form and in mutual agreement.

PROVISION OF EQUIPMENT AND MATERIALS BY CUSTOMER

All tooling, work holding equipment, parts, etc. are to be supplied with DDP supplying plant EMAG Incoterms 2020. We wish to point out that we accept no responsibility for the functional capability and working order of any equipment provided.

The functional capability of the provided equipment is the responsibility of the buyer.

In the event of problems, we assure you of our constructive collaboration. Any additional costs arising from modification, optimization, etc. will be charged to the customer. Furthermore, we do not guarantee any quoted machining times in such cases.

When orders cannot be carried out on schedule, caused by material supplied by the customer or if technical problems should occur, the machine will be invoiced on the originally agreed delivery date, when the relevant payment will become due.

The agreed upon date of delivery date may be postponed to the time to achieve full operability, which shall not result in any event of default of the manufacturer.

DELIVERY PERIOD

Approx. 12 months FOB German Seaport when order will be placed until calendar week 34/2022. The delivery period is not binding and can change between the submission of quotation and placement of order.

Please discuss the subject of delivery with us at time of order placement and we will do everything possible to meet your requirements. Changes regarding the component/s or the machines or the order content after order placement may result in changes in price and delivery.

Compliance of delivery date requires timely down and / or partial payment by the customer within the prescribed payment terms. If receipt of down and or partial payment is delayed, the delivery shall be postponed accordingly, which shall not result in any event of default of the manufacturer.

CONFORMITY WITH CUSTOMERS` REGULATIONS

has not been taken into account. Where unavoidable deviations from such regulations prove necessary, specific authorization will be requested.

PRICES

FCA (Free Carrier, Incoterms 2020). Packing, insurance and transport not included.

The payments are to be made without deduction. With delayed payment the relevant laws apply.

VALIDITY OF PRICES

The prices quoted are valid for 90 days from the date of quotation. Orders placed after this period would be subject to a price increase of 0.5 % per month to cover increases in labour and material costs.

This quotation will replace all previous versions.

Price does not include options (*)

Offer : 196564 / 5

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TERMS OF PAYMENT

100 % of invoice value against irrevocable Letter of Credit payable as follows:
30 % of L/C-value after opening of the L/C as advance payment against simple receipt.
70 % of L/C-value at sight against shipping documents.
All L/C-charges are for buyer's account. L/C must be opened latest 30 days after order confirmation

RETENTION OF TITLE

The machine will remain the property of the builder until complete payment is received from the buyer. Upon placement of order the extended retention of title according to clause 48 of our General Conditions for Sales and Deliveries shall be deemed as agreed.

LIMITATION OF LIABILITY

The liability of the supplier is excluded for indirect damage, defects and consequential damage, or losses, e.g., loss of revenues, downtimes, production stoppages, capital costs, or costs incurred due to operational interruptions. This does not apply to damage willfully caused by the supplier, or in cases of statutory liability, in particular according to the Product Liability Act.

PERIOD OF LIABILITY

Any rights and claims of the buyer caused on defects shall be subject to a limitation period of twelve months. The limitation period starts with the final acceptance, but no later than 3 months from delivery, and does not apply to wear and tear parts.

WORK PIECES FOR MACHINE ACCEPTANCE

Both, pre-acceptance and final acceptance tests are carried out with either an EMAG test component to VDI standard, or the work piece specified by the customer's order.

The customer will supply an agreed number of work pieces for machining process optimisation and pre-acceptance tests. These work pieces must be from a single batch and of a quality that corresponds with the relevant manufacturing drawing. For preparatory work and preliminary machine acceptance the customer will, at the agreed time, supply EMAG free of charge with a sufficient number of sample work pieces of a quality that corresponds with the relevant manufacturing drawing. Any delay in the supply of these work pieces will result in the machine delivery being delayed and the starting date of a possible delayed machine delivery penalty being deferred accordingly.

FINAL ACCEPTANCE AND FINAL PAYMENT

Final acceptance tests are carried out as soon as the machine has been installed and commissioned. For this purpose, the buyer has to provide a sufficient number of work pieces, personnel and capacity to measure and evaluate the acceptance components. Once the machine is taken into production and the agreed component quality and cycle time have been achieved, final acceptance is considered to have taken place. Where final acceptance of the machine is not possible owing to reasons beyond EMAG's control (missing acceptance components, etc.), final contractual payment falls due immediately.

Price does not include options (*)

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TECHNICAL AVAILABILITY

Our confirmation of the technical availability according to VDI 3423 is based on following conditions:
Exclusively qualified operating personnel, monthly availability protocols addressed to our service manager, demonstrable preventive maintenance is carried out according to our machine documentation, connection to our tele service system if requested by our service department.
In case of non-compliance of these conditions, our commitment expires.

TECHNICAL DEVELOPMENT

In the interest of technical progress, we reserve ourselves the right to make design changes which are due to continuous product development.

RESERVATION CLAUSE

For offers within the European Union (without Germany):

„Notification of Intra-Community delivery according to Art. 22 (10) EC Dual-Use Regulation:

The following purchased goods are subject to export controls. For the export of these goods from the Community an export license is necessary.”

For offers outside the European Union:

It is agreed, that the legally binding conclusion of this offer is subject to the suspensive condition that the competent authority grant the necessary export license.

ADDITIONAL PROVISIONS

In addition and subordinated to this quotation our General Conditions for Sales and Deliveries for International Business shall apply.

http://www.emag.com/Delivery_conditions_UN.pdf

<https://www.emag.com/de/iot-services.html>

<https://www.emag.com/media-recommendation/>

This document was created electronically and is valid without signature.

Sincerely,

EMAG Maschinenfabrik GmbH

Hartmut von Fugler
Managing Director

Do-Youn Koo
Area Sales Manager

Enclosures

EMAG-General conditions for sales and deliveries (UN)

Layout

Schematic of machining, TImestudy, Quality commitments

Part MAKER List Hyundai WIA Mexico Commented by EMAG

STN KV EI + Mech + Fluid HYUNDAI WIA_DG_V ; EMAG-Interface to external handling systems

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Commerzbank AG Tübingen IBAN DE09 6414 0036 0890 9103 00 BIC COBADEFF641
Kreissparkasse Göppingen IBAN DE96 6105 0000 0015 5885 93 BIC GOPSDE6GXXX
Bayerische Landesbank München IBAN DE59 7005 0000 0004 2569 67 BIC BYLADEMMXXX

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